

Deep Learning for Computer Vision

Lecture 5: Semantic Segmentation

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Exercise Presentation

- Decide beforehand who presents which part.
- Prepare so that you can present the whole exercise in ca. 15 minutes.
- **We have tutorials this week where you present exercise 2**

Technical Issues with the Exercises

Please ask in the chat:

- Either in the general chat, the chat of your tutorial, or your tutor
- Don't ask me; the tutors are taking care of the technical issues / support with the exercises

Reminder: Sign up on FlexNow for the exercises

- Latest date to sign up: 12.12.25
- You cannot get credits if you have not signed up by this date!

Semester: WS25/26 ▾

☐ nur Prüfungen mit Prüfungsterminen im gewählten Semester

☐ nur Prüfungen mit fehlenden Bewertungen

Prüfung: B.Inf.1237.Ue: Deep Learning for Computer Vision (Exercise) [Anzahl TN: 74/0/0] ▾

Prüfungstermin:

 LV-Nr. 503095; PAPE - 12.12.2025 - Deep Learning for Computer Vision [Anzahl TN: 74/0/0]

Next exercise

- Transfer learning for medical image classification
 - Using ImageNet and a domain specific pre-training dataset
- Challenge: improve your model using techniques we discussed in the lecture (data augmentation, model ensembling, ...)
 - Submit predictions of your best model on a hold-out test set
 - Best group will get a small price ;)
- Already available on Stud.IP, how to submit the predictions is explained in the exercise.

STARTUP COMPETITION LIFT-OFF 2026



WHY YOU SHOULD PARTICIPATE

HELP WITH THE DEVELOPMENT AND ELABORATION OF YOUR BUSINESS MODEL
FREE PARTICIPATION IN A WIDE RANGE OF WORKSHOPS
ATTRACTIVE CASH AND NON-CASH PRIZES
PR EFFECTS & NETWORKING
PARTICIPATION IN THE EXCLUSIVE MENTORING PROGRAMME
ACQUIRE FIRST CUSTOMERS
PITCH TRAINING



SIGN UP NOW!



Agenda for today

Computer vision tasks: beyond image classification

Semantic segmentation

- Datasets
- Architectures
- Loss functions

Computer Vision Tasks

IMAGENET

Dataset and benchmark on image classification

- 1 million images with ground truth class labels for training (hand-annotated)
- 1000 object categories



IMGENET

- ILSVRC Challenge: compare models on ImageNET
 - From 2010-2017
 - 2012: AlexNet wins and starts current DL boom
 - Defining challenge of modern CV & AI
- Today: “default” dataset for new image classification architectures
- BUT: performance is saturated
 - Remaining error rate similar to label error frequency of the data
 - Better-than-human-performance (but CV is far from solved!)



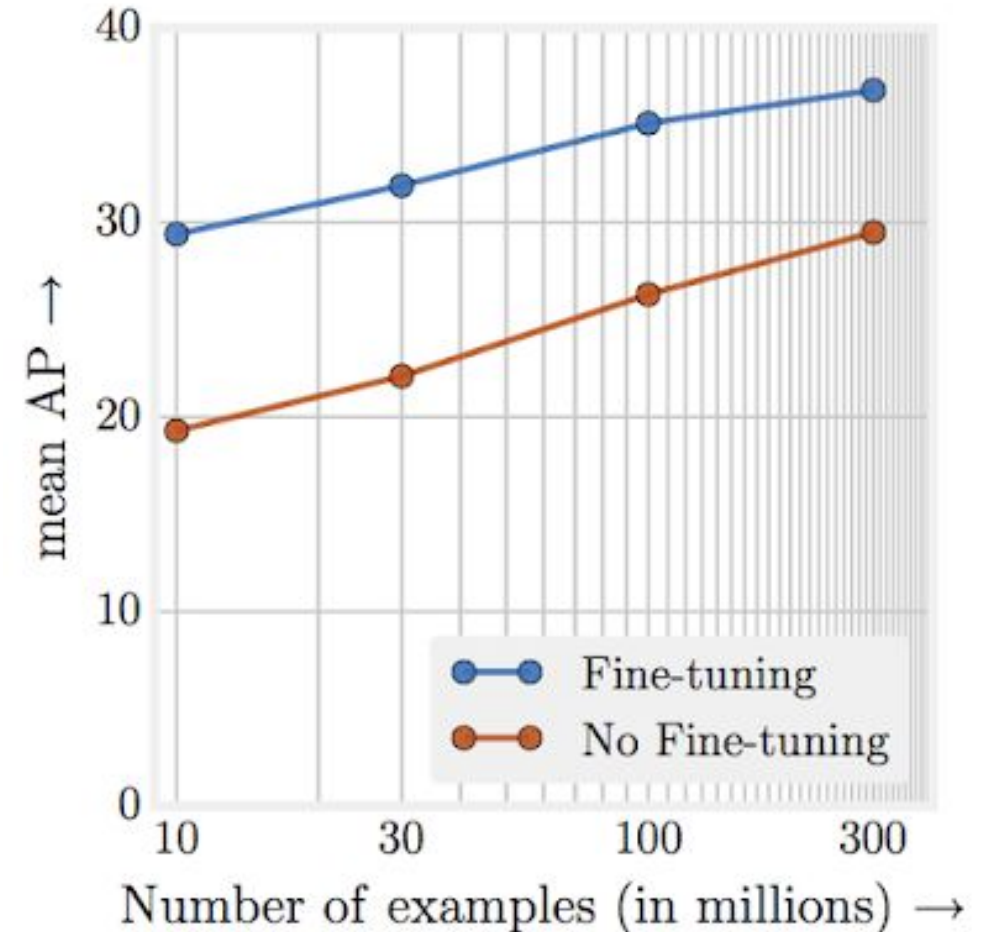
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Tasks and challenges drive progress in CV.
We need better datasets, tasks & challenges!

Bigger than IMGENET ?

JFT-300M: Internal Google Dataset:

- 300 Million images with 1 Billion labels (multiple labels per image possible)
- Generated in a semi-automated way, aggregating labels from several sources
- Main finding: performance increases log. with dataset size
- Similar internal datasets at Facebook etc.



<https://ai.googleblog.com/2017/07/revisiting-unreasonable-effectiveness.html>

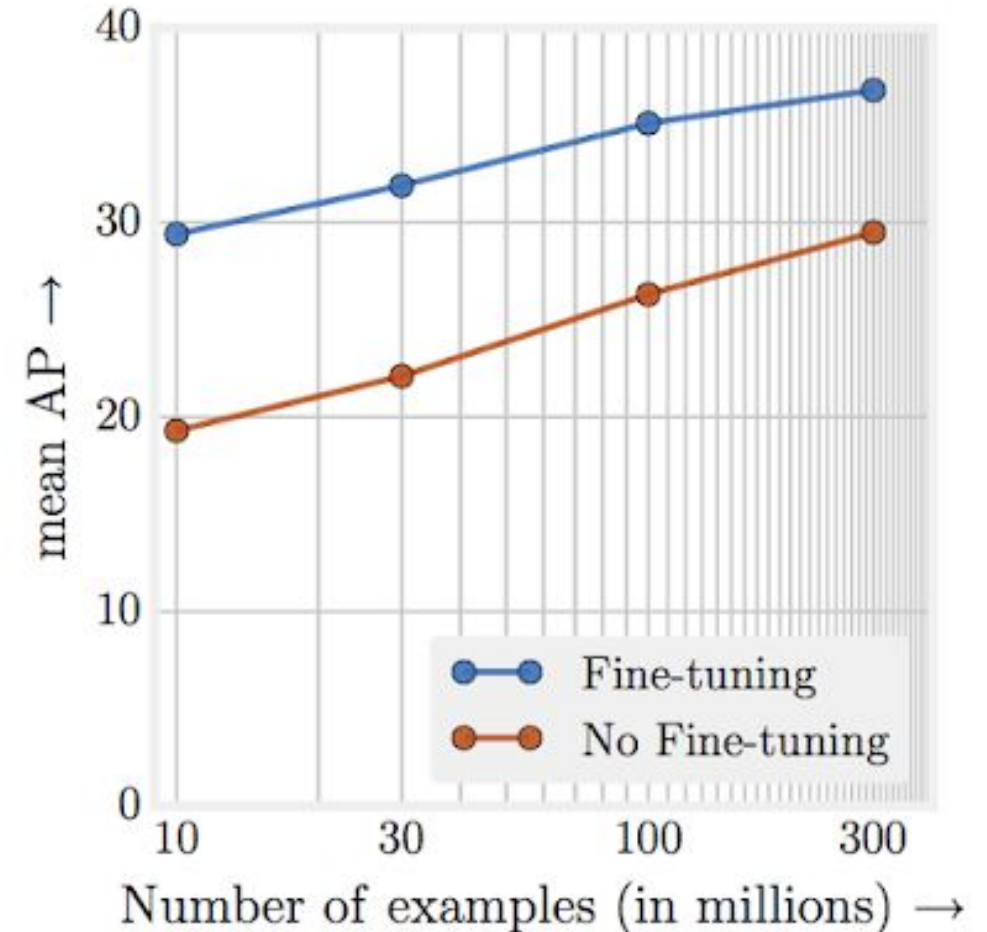
https://openaccess.thecvf.com/content_ICCV_2017/papers/Sun_Revisiting_Unreasonable_Effectiveness_ICCV_2017_paper.pdf

Bigger than IMGENET ?

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- 300 Million images with 1 Billion labels (multiple labels per image possible)
- Generated in a semi-automated way, aggregating labels from several sources
- Main finding: performance increases log. with dataset size
- Similar internal datasets at Facebook etc.

BUT: data not published, massive resources needed to collect or even train on such datasets!



Beyond Image Classification

How to label the image on the right?

- Single label
 - Pedestrian?
 - Car?
 - House?



Beyond Image Classification

How to label the image on the right?

- Single label
 - Pedestrian?
 - Car?
 - House?
- Multiple labels
 - Pedestrian, Car, House, Skyscraper, ...

Hopeless!



Beyond Image Classification

How to label the image on the right?

- Single label
 - Pedestrian?
 - Car?
 - House?
- Multiple labels
 - Pedestrian, Car, House, Skyscraper, ...

Hopeless!

What for?

(Human) Vision is much more complex than classification!



Beyond Image Classification

- Last lecture: Transfer learning
 - Pretrain model on large classification dataset (ImageNet)
 - Finetune to solve classification on much smaller dataset
 - **or to solve a different vision task**
- This week and next weeks: More complex vision tasks
 - Semantic segmentation
 - Object detection
 - Instance segmentation
 - Image synthesis
 - ...

Vision Tasks

- So far: image classification
 - Limitations: does not require understanding of spatial relations in image
- “Scene parsing”: semantic segmentation, instance segmentation, object detection, ...
 - Location and relation of objects in images matters!
- Image-to-image tasks: colorization, denoising, ...
 - Transform image into different (but related) image
- Image synthesis: style transfer, image generation (from caption), ...
- etc.

Scene parsing

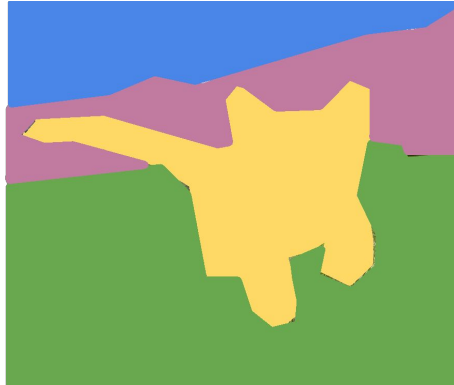
Image
Classification



Cat

No spatial extent

Semantic
Segmentation



Sky, Trees, Grass, Cat

No objects, just pixels

Object
Detection



Dog1, Dog2, Cat1

Multiple objects

Instance
Segmentation



Dog1, Dog2, Cat1

Image-to-image tasks

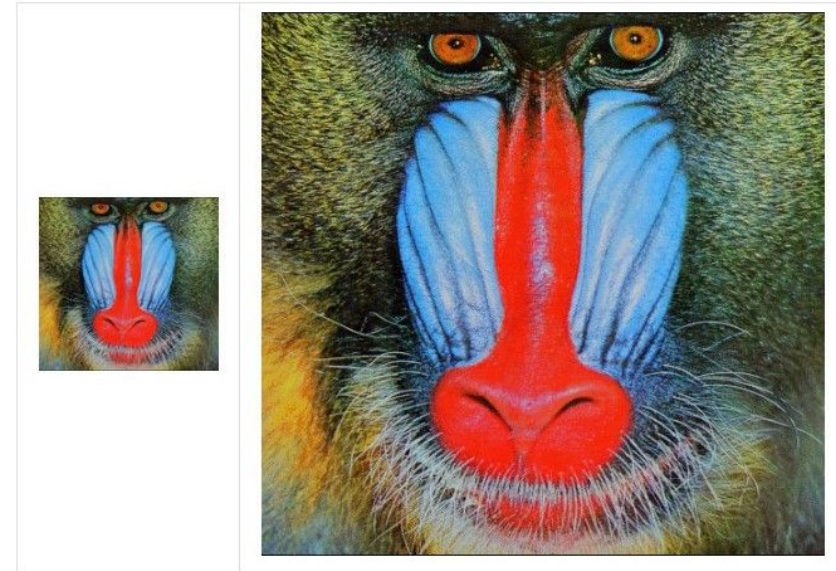


Colorization

Image-to-image tasks



Colorization

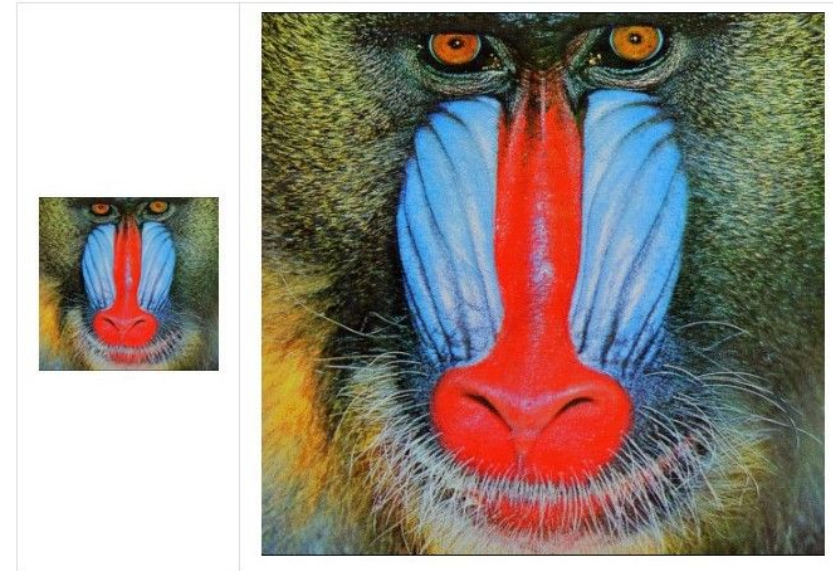


Super-resolution

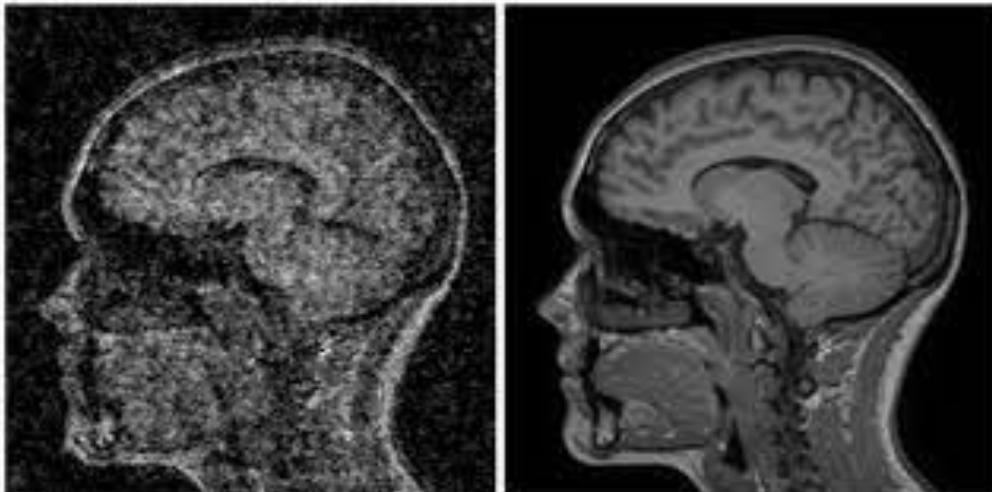
Image-to-image tasks



Colorization



Super-resolution



Denoising

Image synthesis



Content Image



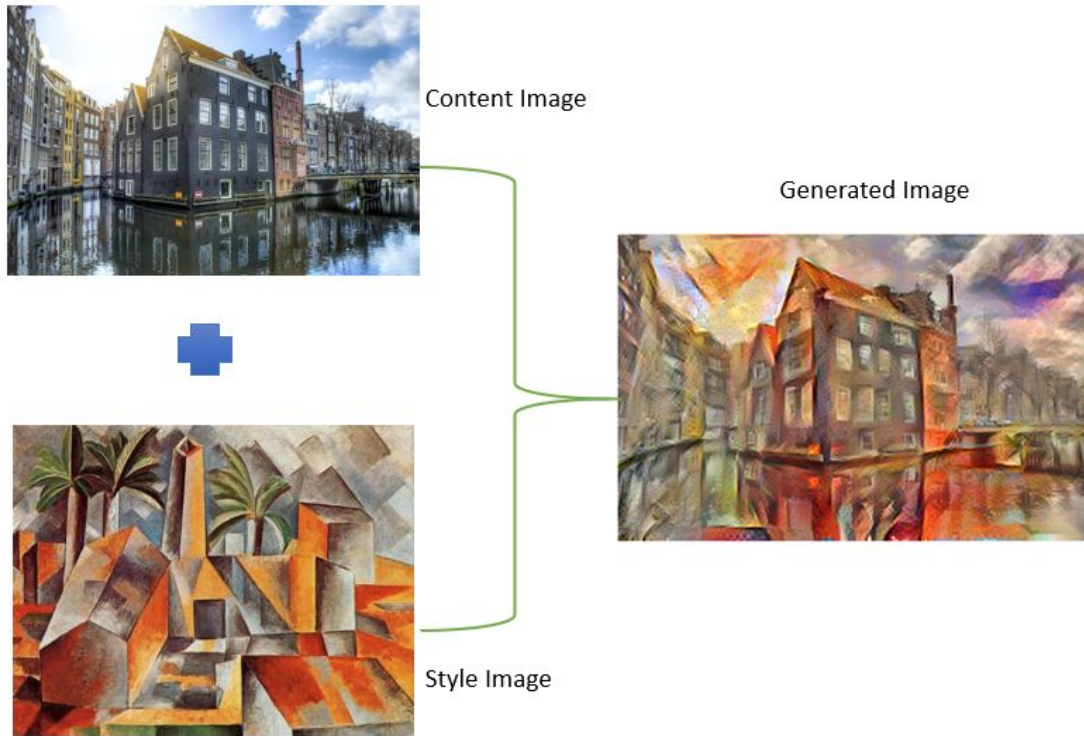
Style Image



Generated Image

Style Transfer

Image synthesis



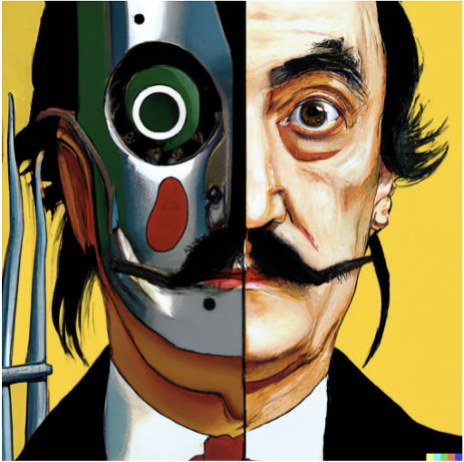
Style Transfer



Image generation

Image synthesis

Caption based image generation



vibrant portrait painting of Salvador Dalí with a robotic half face



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it



an espresso machine that makes coffee from human souls, artstation

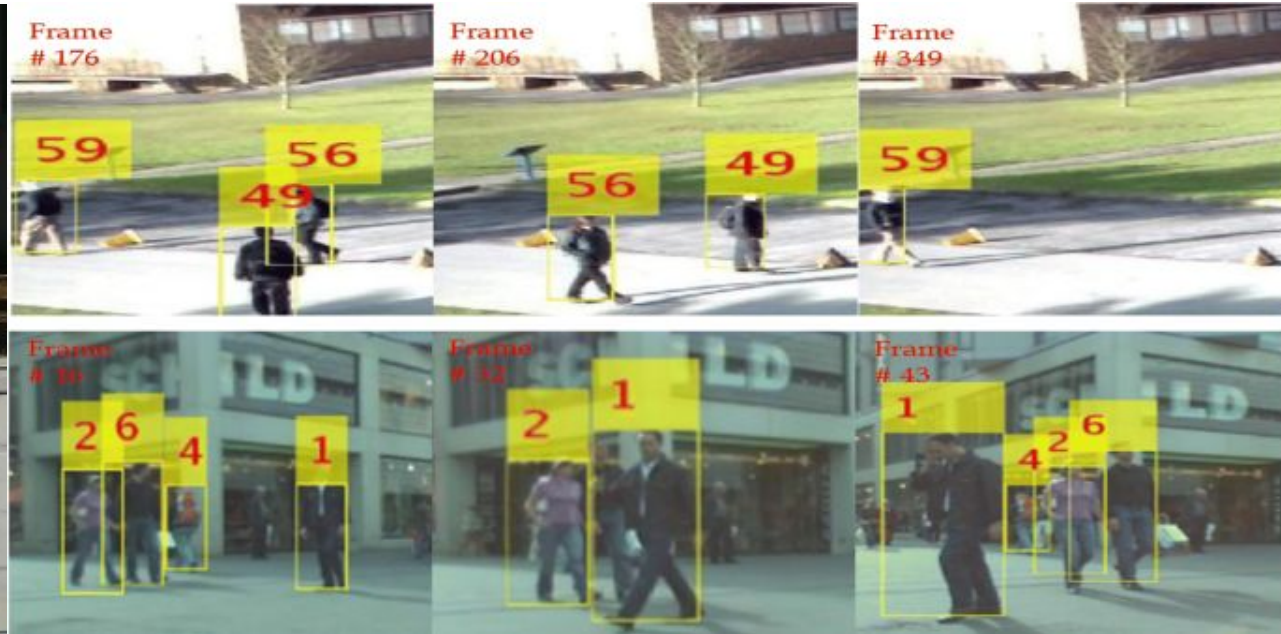
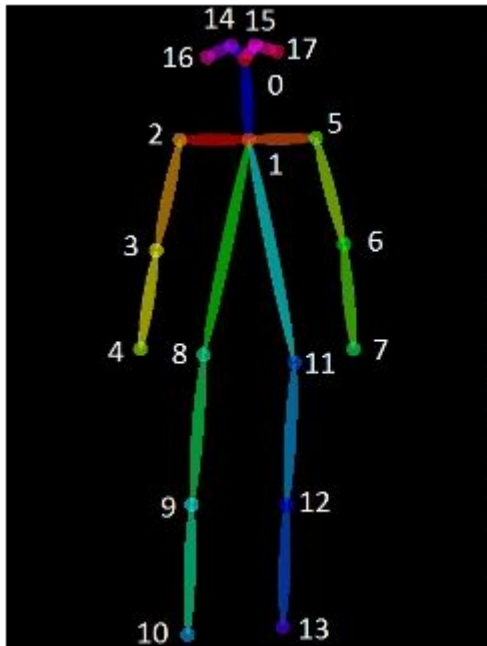


panda mad scientist mixing sparkling chemicals, artstation



a corgi's head depicted as an explosion of a nebula

And more...



Human Pose Estimation

Pedestrian Tracking (Video!)

-> Composite tasks!

Semantic Segmentation

Tasks & Datasets

Segmentation tasks

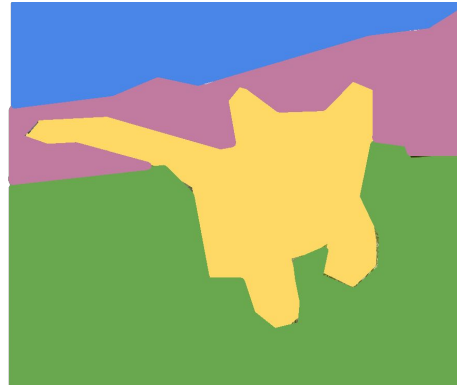
Image
Classification



Cat

No spatial extent

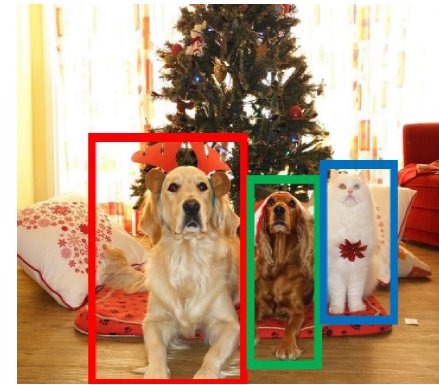
Semantic
Segmentation



Sky, Trees, Grass, Cat

No objects, just pixels

Object
Detection



Dog, Dog, Cat

Multiple objects

Instance
Segmentation



Dog, Dog, Cat

Segmentation tasks

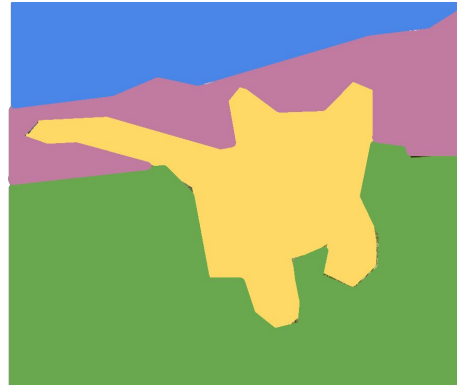
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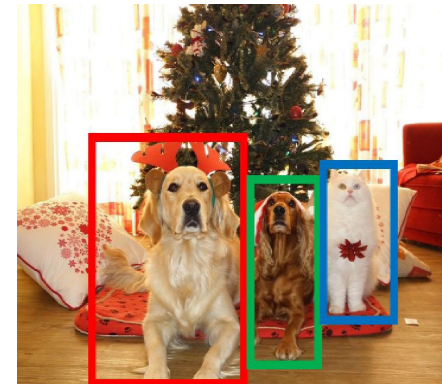
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Segmentation



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Segmentation



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Multiple objects

Tasks & Datasets

- Natural images: semantic classes (as in image classification):
 - “cat”, “dog”, “car”, “person”, ...

Tasks & Datasets

- Natural images: semantic classes (as in image classification):
 - “cat”, “dog”, “car”, “person”, ...
- Biomedical images: structures of interest in the image
 - radiology: lesions in MRI images, ...
 - light microscopy: nuclei in microscopy images of cell cultures, ...
 - electron microscopy: organelles (membrane, golgi, etc.) in cells
- Material sciences: defects in materials, e.g. imaged in x-ray or electron microscopy
- etc.

Segmentation in natural images

Pascal VOC (visual object classes):

- **semantic segmentation**, object detection & instance segmentation
- 20 classes
- 11,530 images

<http://host.robots.ox.ac.uk/pascal/VOC>



bus



train



person



food



cow



bird



sheep



boat



horse



dog

Segmentation in natural images

Pascal VOC (visual object classes):

- **semantic segmentation**, object detection & instance segmentation
- 20 classes
- 11,530 images

Similar, but larger dataset:
COCO (Common Objects in Context)
330k images, 80 classes
<https://cocodataset.org/>



bus



train



person



food



cow



bird



sheep



boat



horse



dog

Segmentation in natural images

Cityscapes: segmentation in traffic scenes

- **semantic segmentation**,
object detection &
instance segmentation
- 30 classes
- 25,000 images

<https://www.cityscapes-dataset.com/>

Segmentation in natural images

<https://www.cityscapes-dataset.com/>



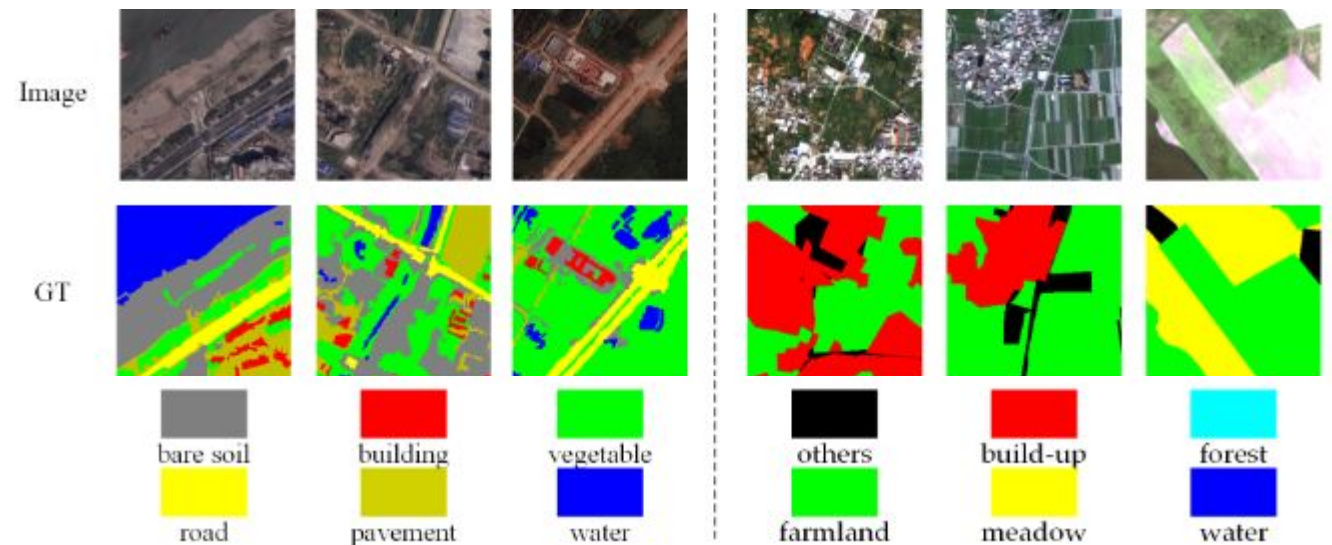
Segmentation in natural images

Image from

<https://arxiv.org/pdf/2007.13083.pdf>

And many more applications, e.g.:

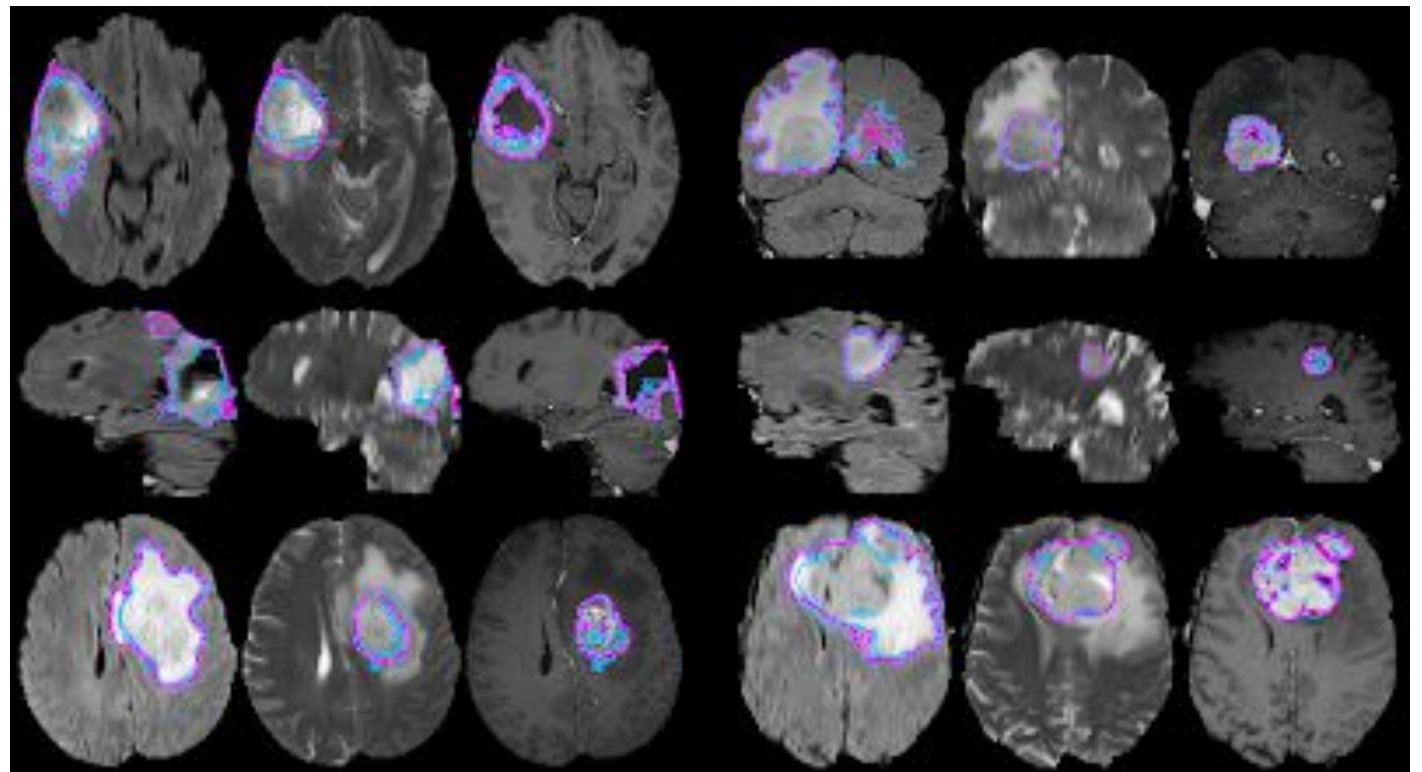
- Segmentation for remote sensing (satellite or drone imagery)
- Segmentation for industry
 - process automation, finding defects, ...
- ...



Segmentation in biomedical images

BRATS: Brain tumor segmentation in MRI scans

- semantic segmentation
- 4 classes
- 8,000 MRI scans

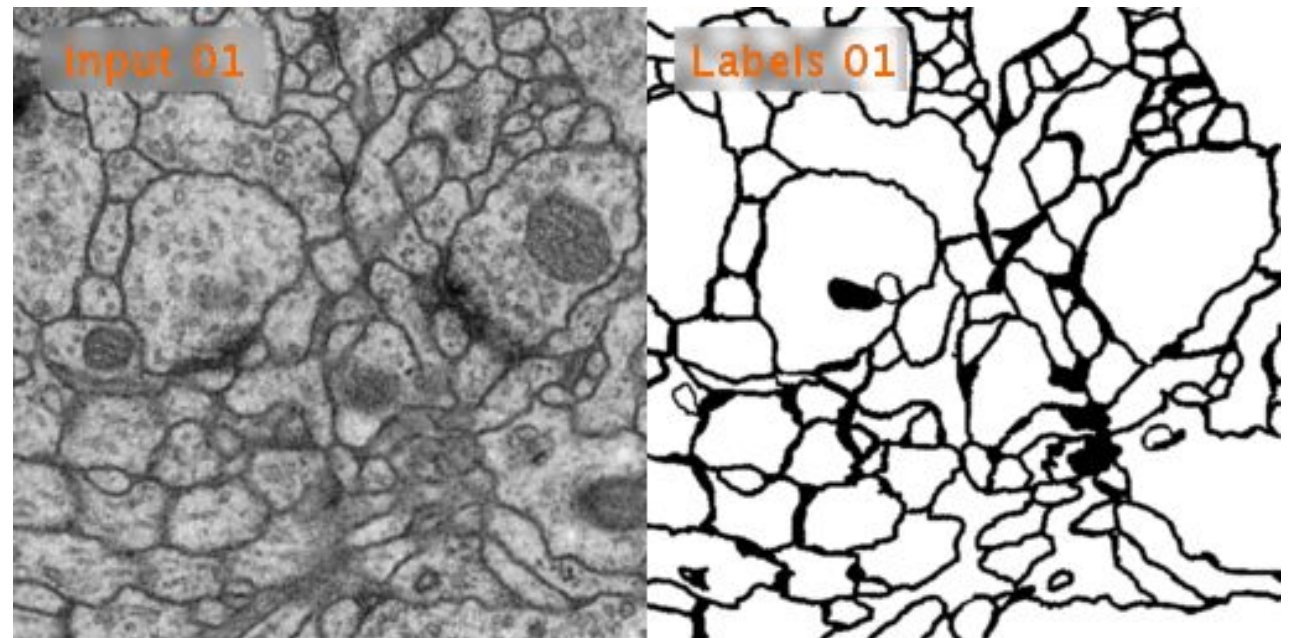


<http://braintumorsegmentation.org/>

Segmentation in biomedical images

Microscopy: many different segmentation tasks

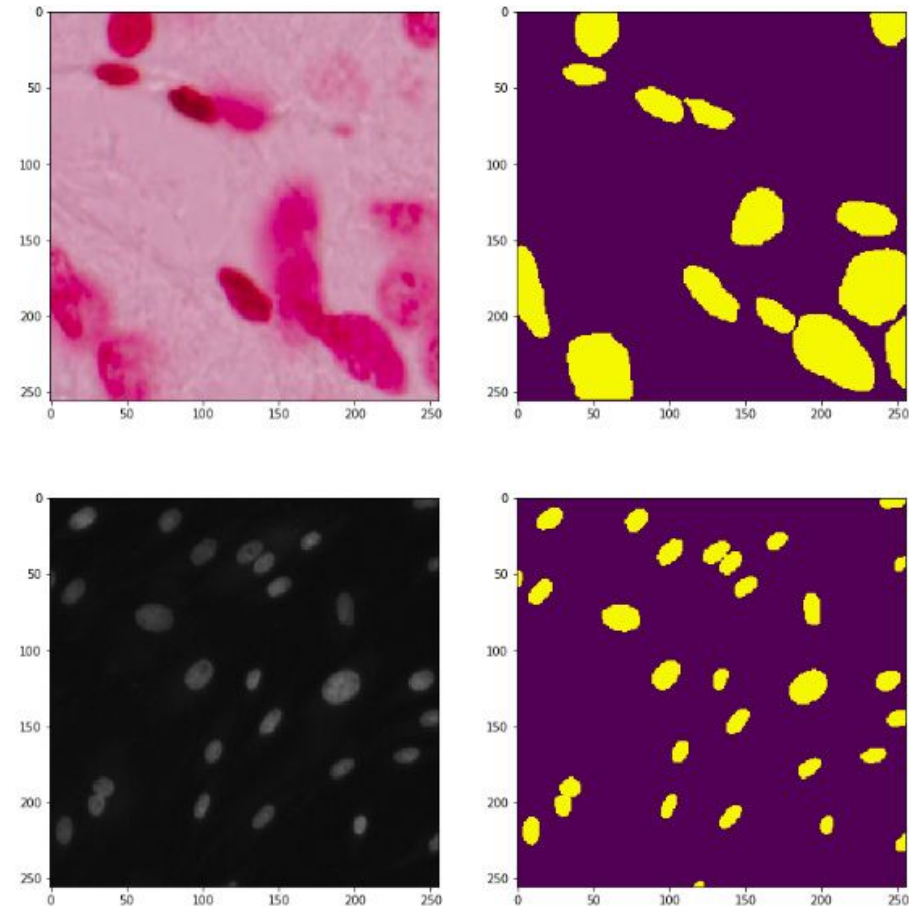
- Neuron segmentation in electron microscopy
<https://imagej.net/events/isbi-2012-segmentation-challenge>



Segmentation in biomedical images

Microscopy: many different segmentation tasks

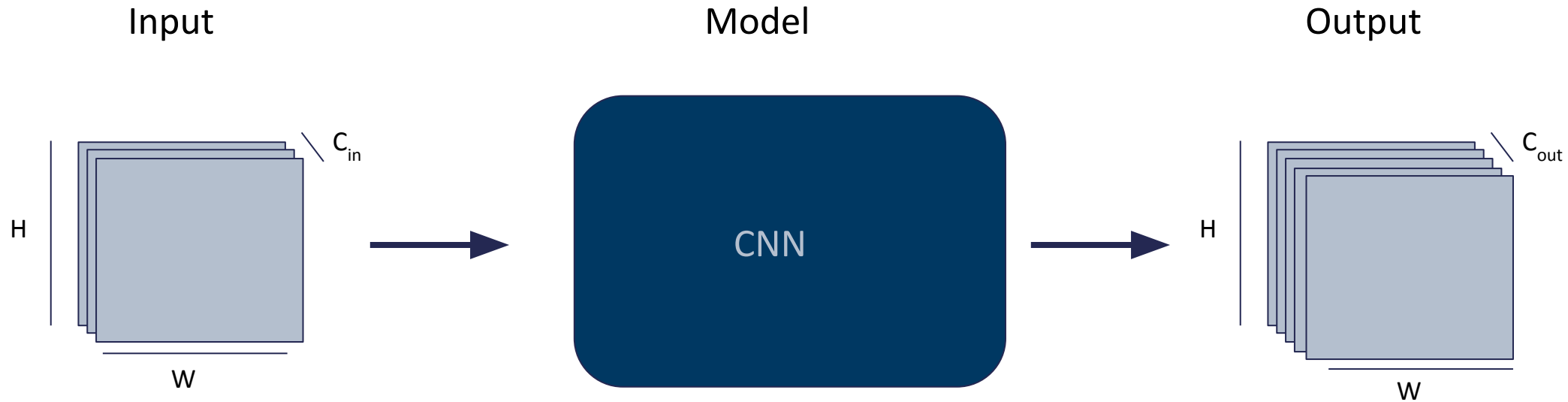
- Neuron segmentation in electron microscopy
<https://imagej.net/events/isbi-2012-segmentation-challenge>
- Nucleus segmentation in light microscopy
<https://www.kaggle.com/c/data-science-bowl-2018>



Semantic Segmentation

Architectures

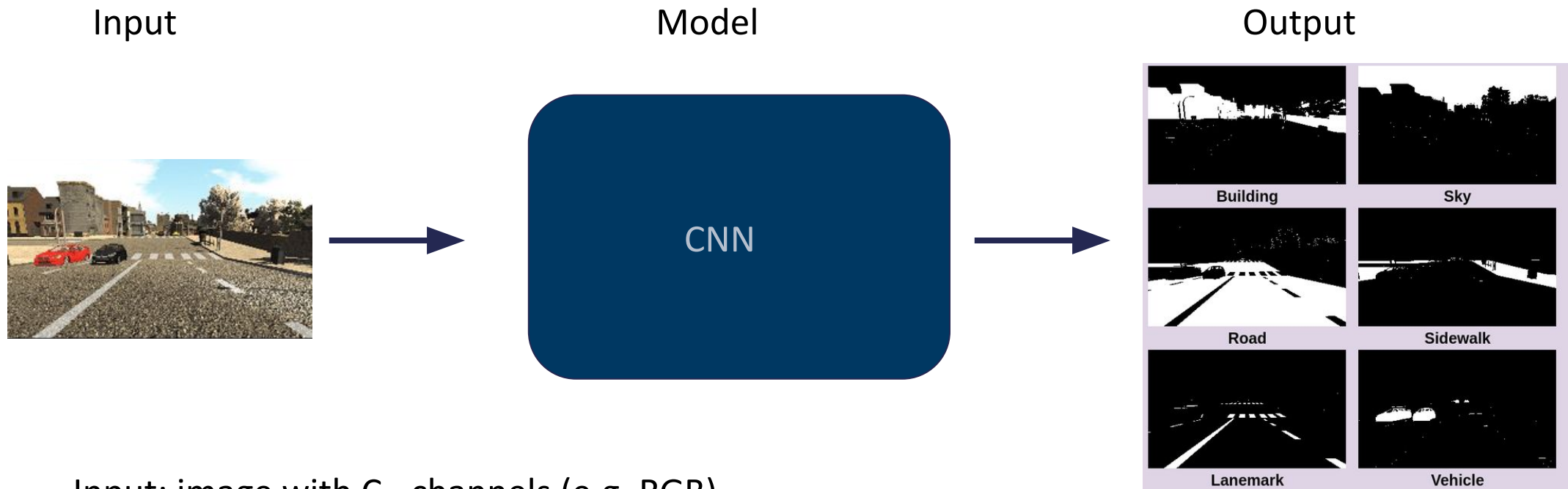
Architectures for semantic segmentation



Input: image with C_{in} channels (e.g. RGB)

Output: image with C_{out} channels, C_{out} = number of classes, one-hot encoding

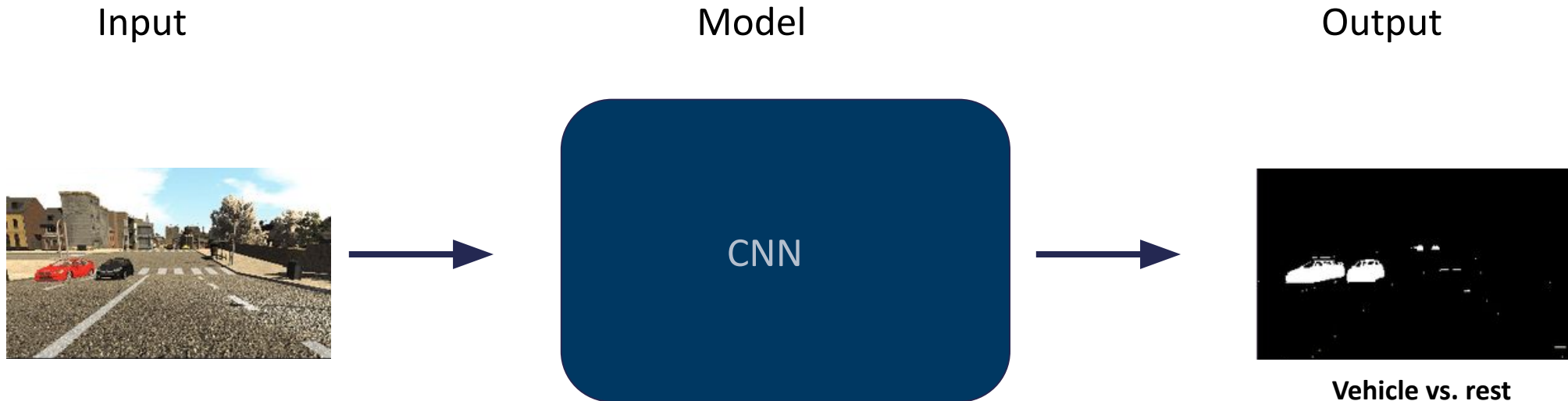
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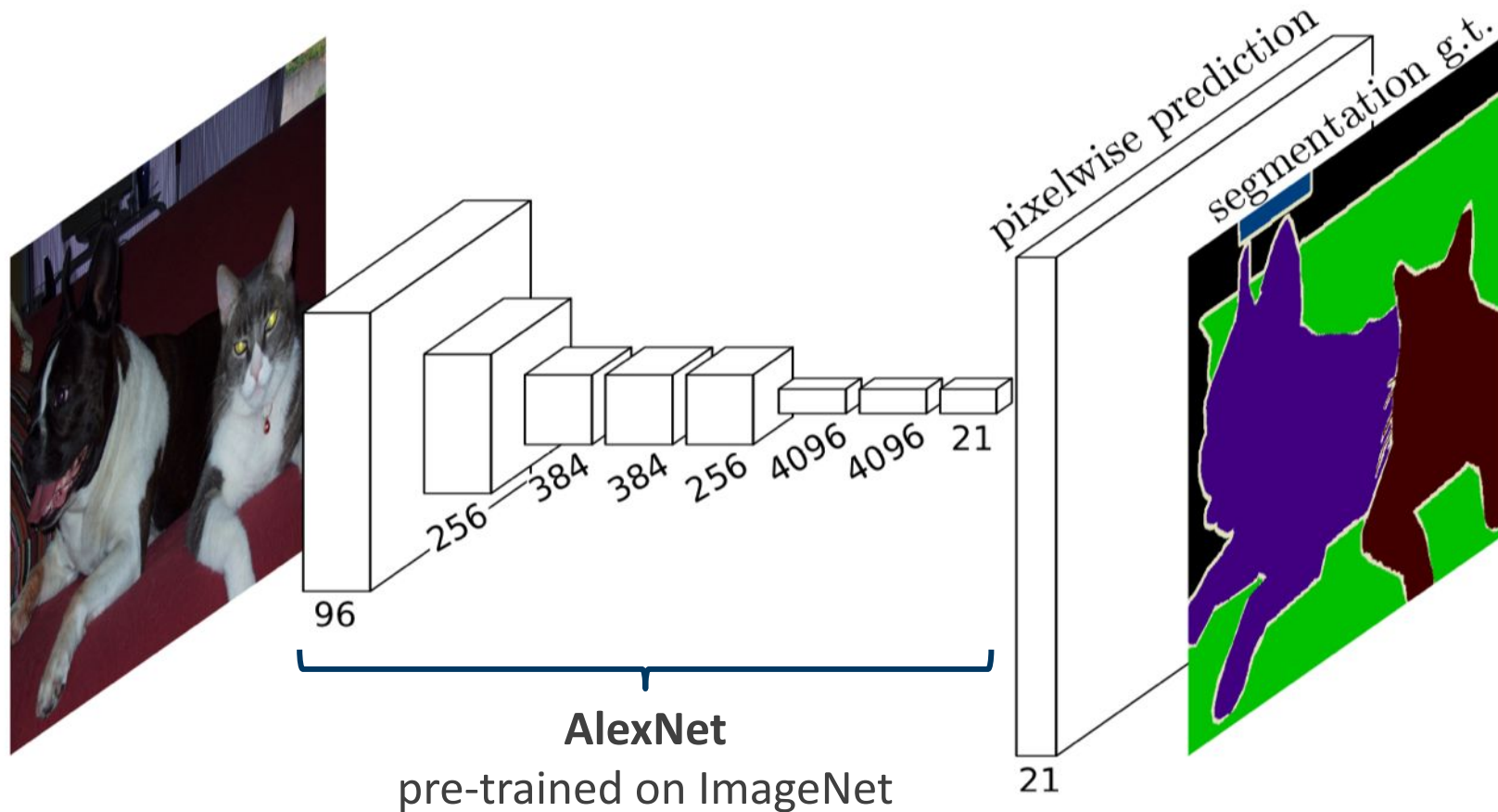
Architectures for semantic segmentation



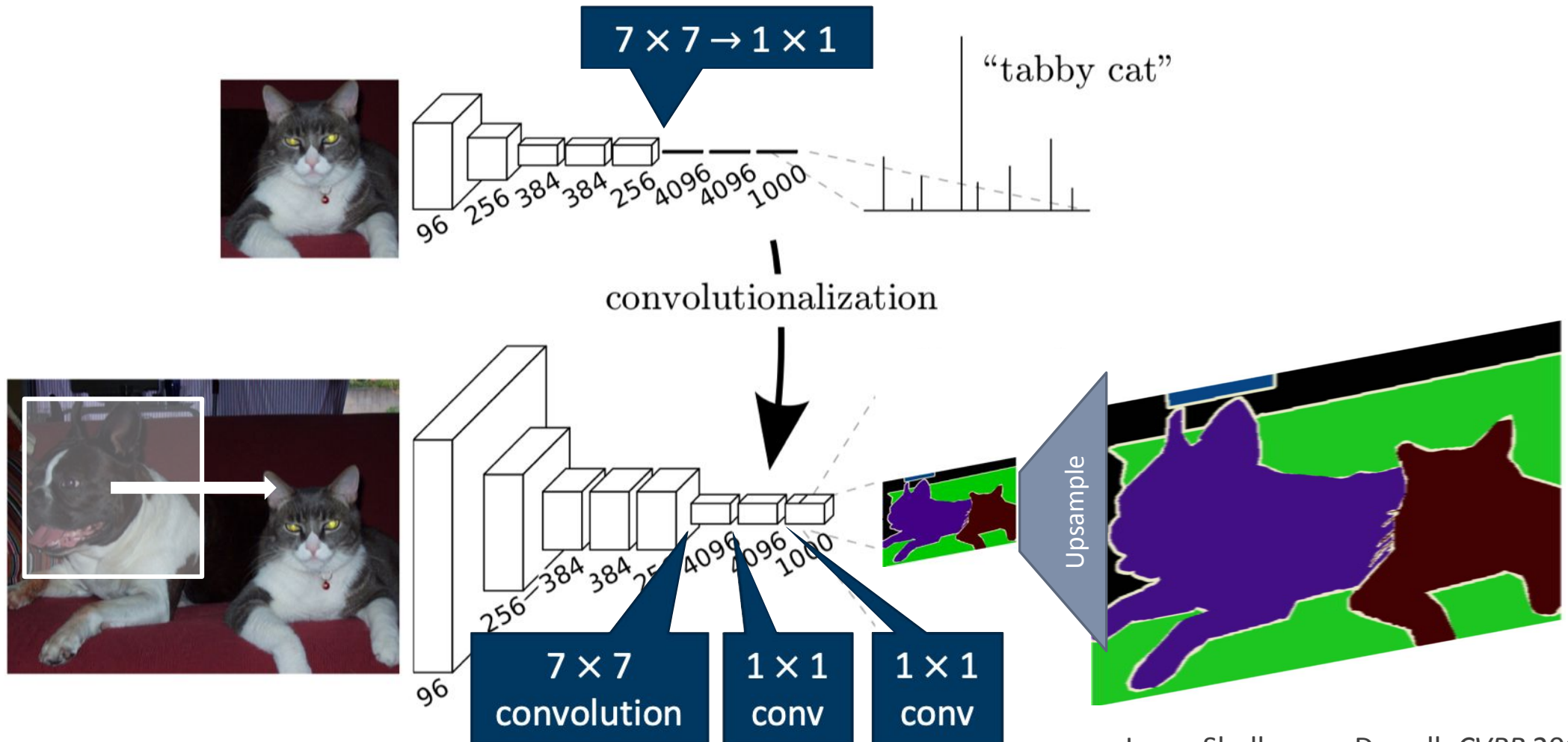
Special case: foreground vs. background segmentation:

$C_{\text{out}} = 1$ (binary segmentation)

Fully convolutional network



Fully convolutional network



Fully convolutional network

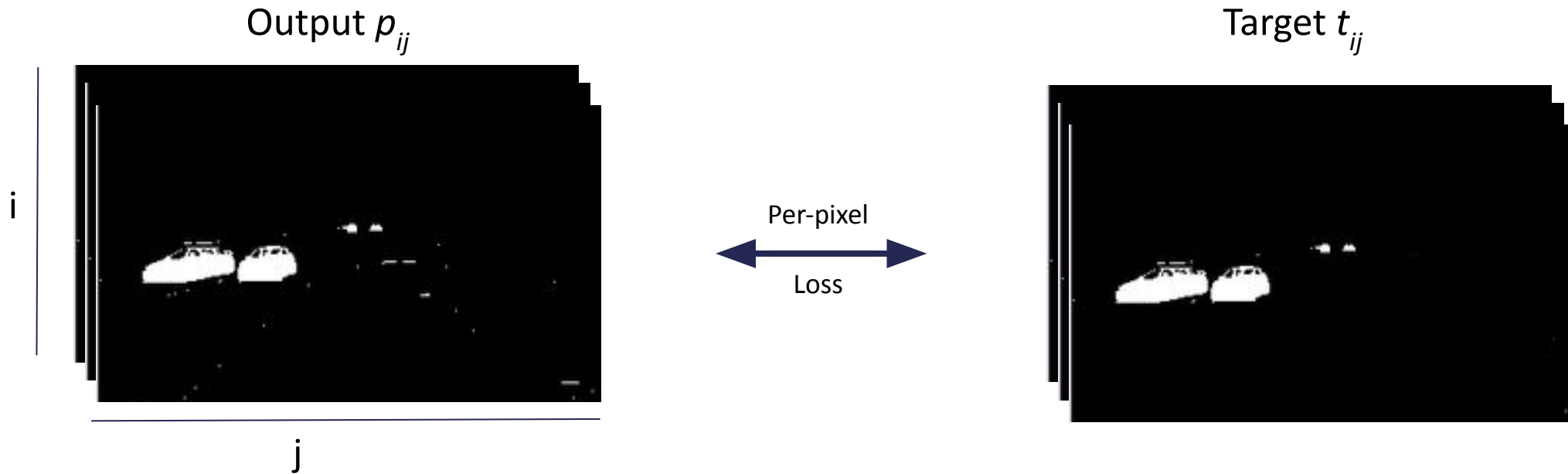
Up to here: have discussed FCN in the last lecture.

More details:

- What loss function is used?
- How does the upsampling work?

Loss function

Pixel-wise cross entropy: $-\frac{1}{N} \sum_{ij} \sum_k \log p_{ij}^k t_{ij}^k$

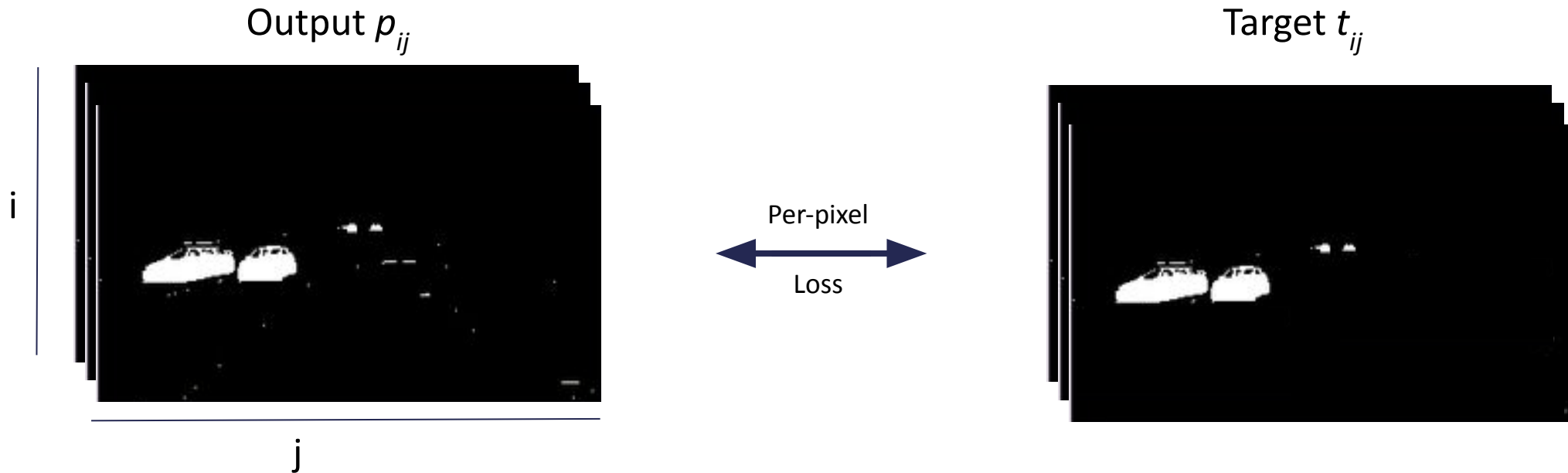


Loss function

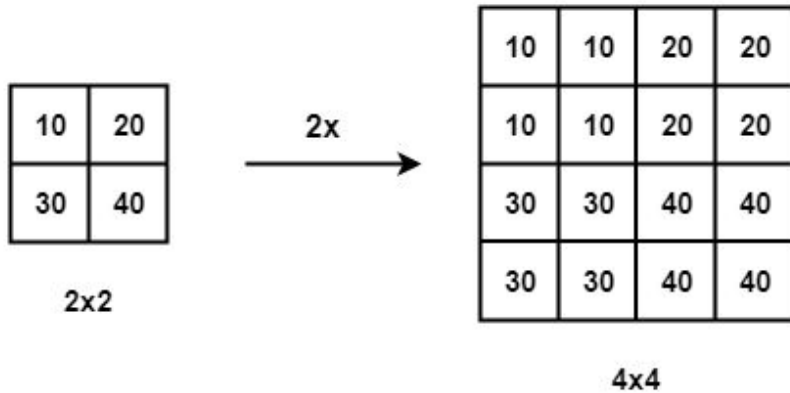
Pixel-wise cross entropy:

$$-\frac{1}{N} \sum_{ij} \sum_k \log p_{ij}^k t_{ij}^k$$

Sum pixels
Sum classes

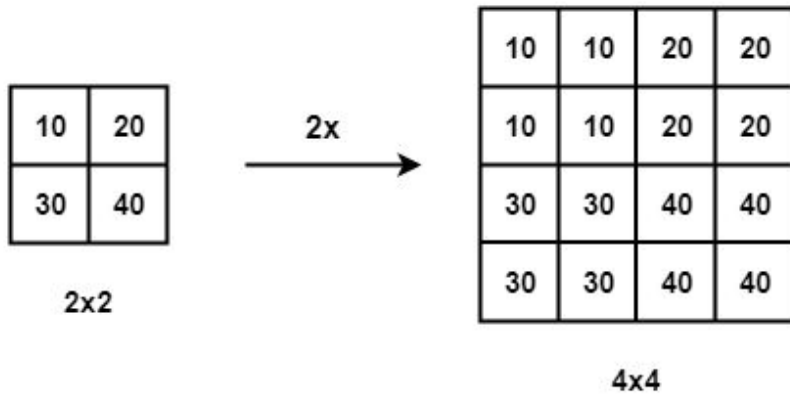


Upsampling



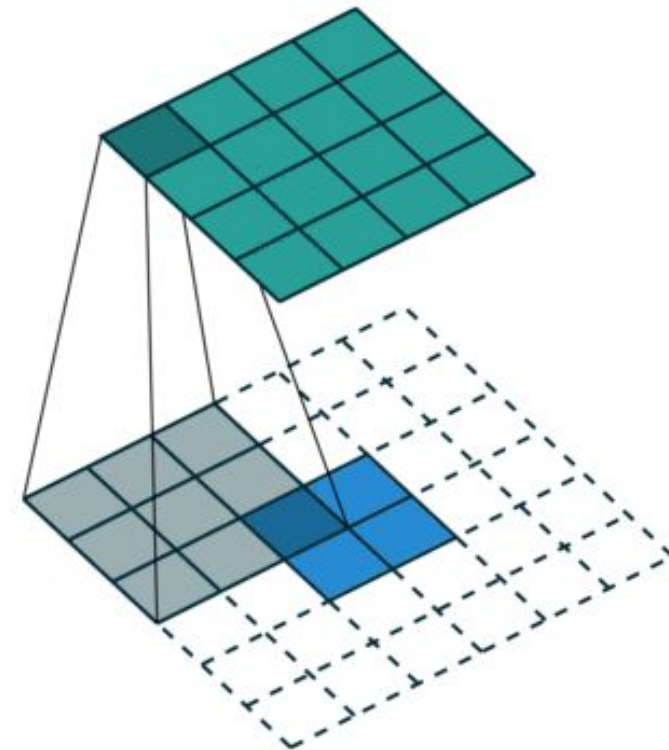
Fixed upsampling: e.g. nearest neighbor or linear interpolation

Upsampling



Fixed upsampling: e.g. nearest neighbor or linear interpolation

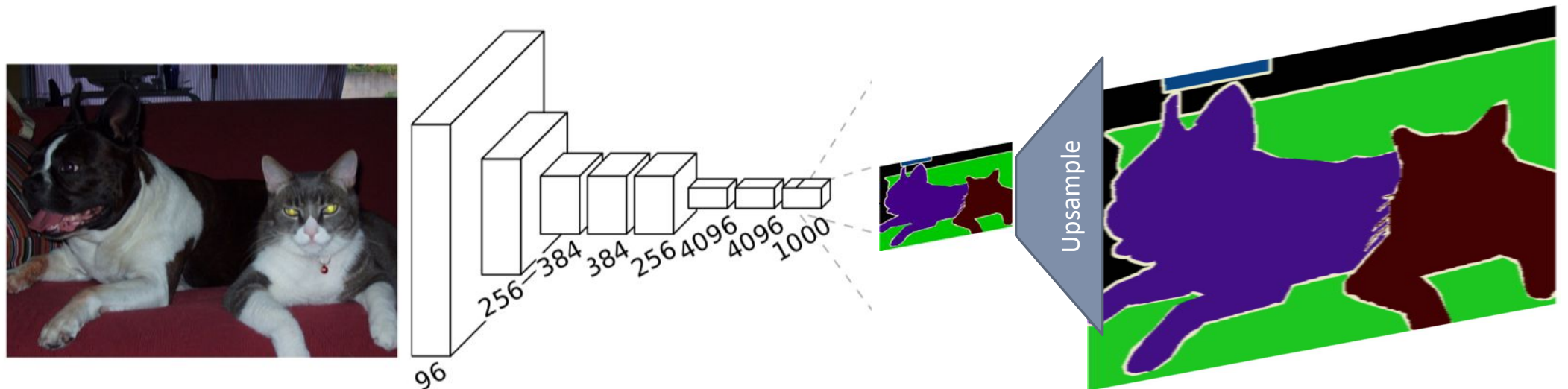
OR



Learned upsampling:
transposed convolutions

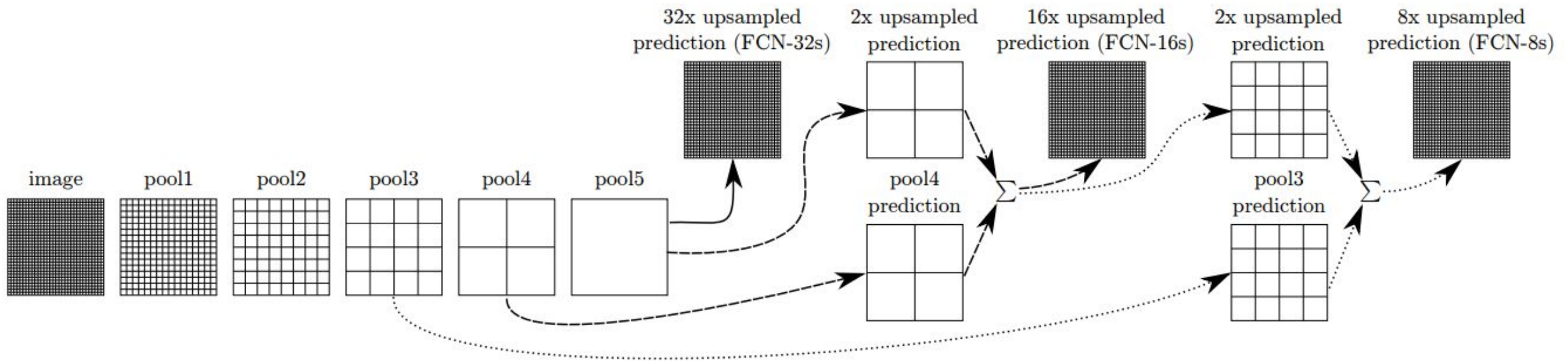
Upsampling

Problem: spatial resolution in last layer is too coarse (10x10 pixels)
-> Will lead to diffuse / blurry segmentation

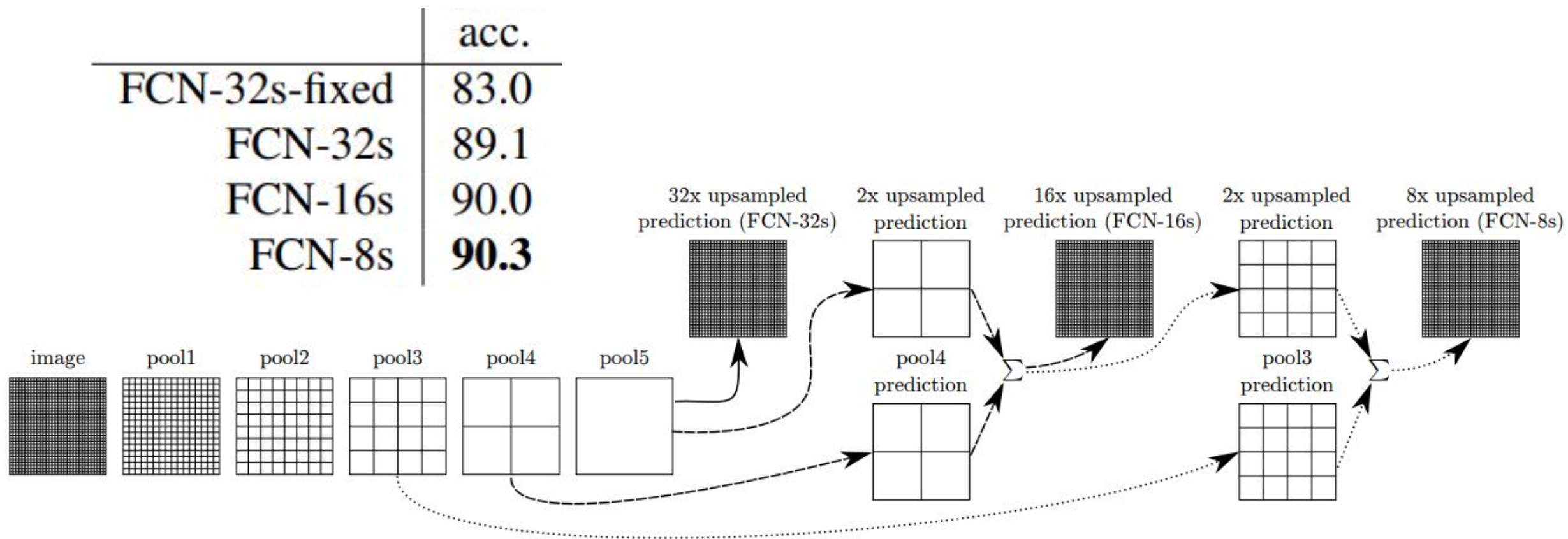


Upsampling

Instead: output of “pool 3, 4, 5” (5 is last pooling), sample to the output size and combine

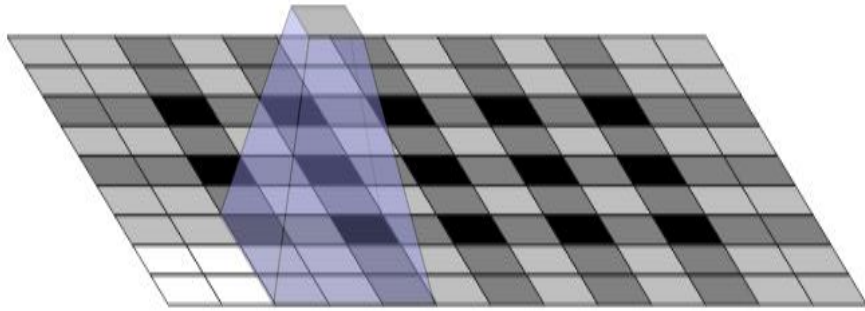


Upsampling

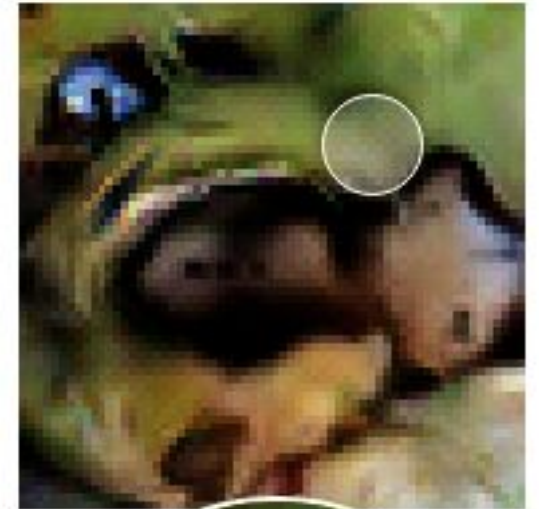


Upsampling

FCN uses transposed convolutions. “Learned upsampling.”
BUT: can lead to checkerboard artifacts.



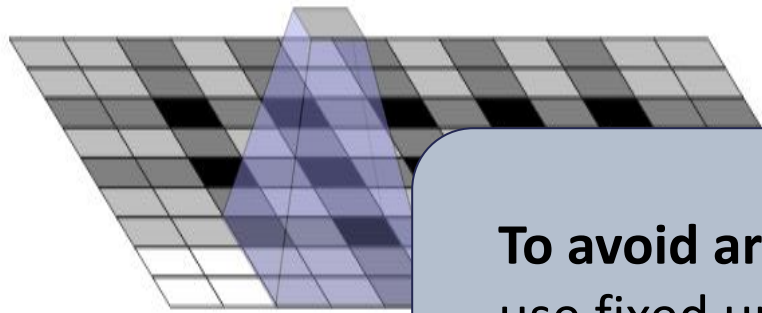
Transposed convolution with “uneven overlap”



distill.pub/2016/deconv-checkerboard

Upsampling

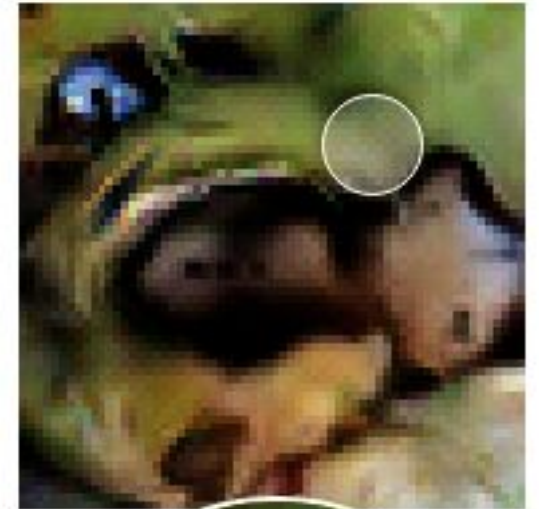
FCN uses transposed convolutions. “Learned upsampling.”
BUT: can lead to checkerboard artifacts



Transposed convol

To avoid artifacts:

use fixed upsampling (nearest neighbor
or linear) followed by 1x1 convolutions



distill.pub/2016/deconv-checkerboard

Architecture improvements

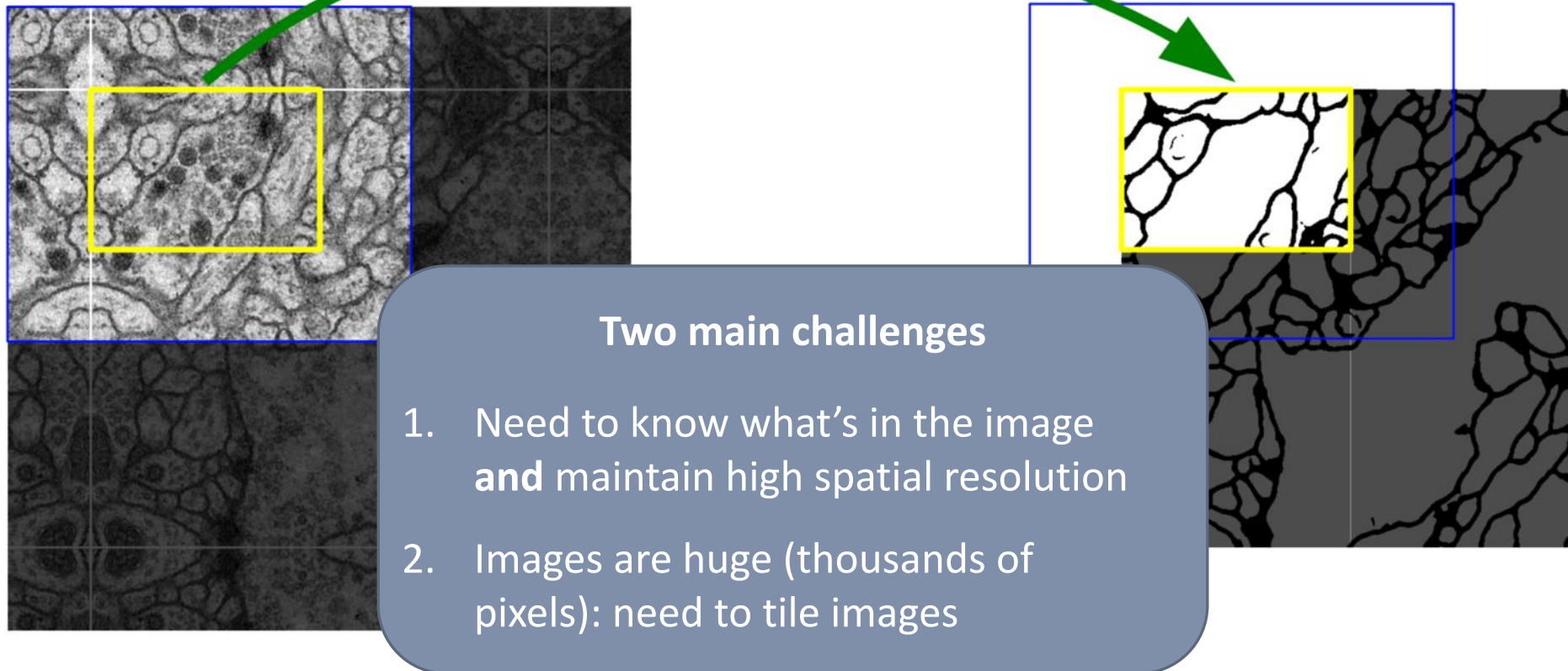
- [Holistically nested edge detection](#): multi-scale input image combination
- [U-Net](#): encoder-decoder structure + skip connections for better spatial resolution
- [DeepLab](#): atrous convolutions and spatial pyramid pooling
- [Hierarchical Multi-Scale Attention](#): multi-scale attention mechanism

Architecture improvements

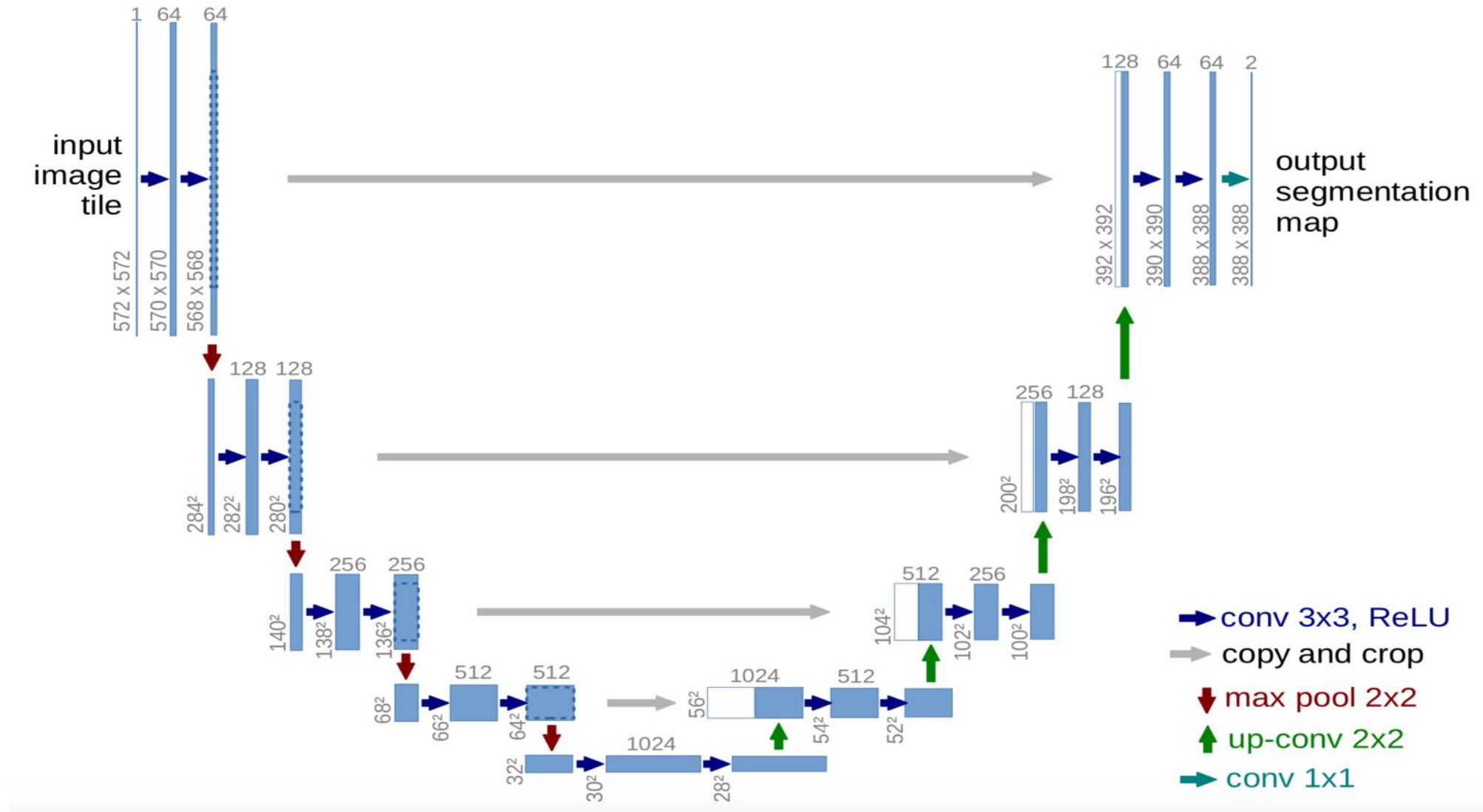
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- [U-Net](#): **encoder-decoder structure + skip connections for better spatial resolution**
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- [Hierarchical Multi-Scale Attention](#): multi-scale attention mechanism

U-net: Biomedical image segmentation

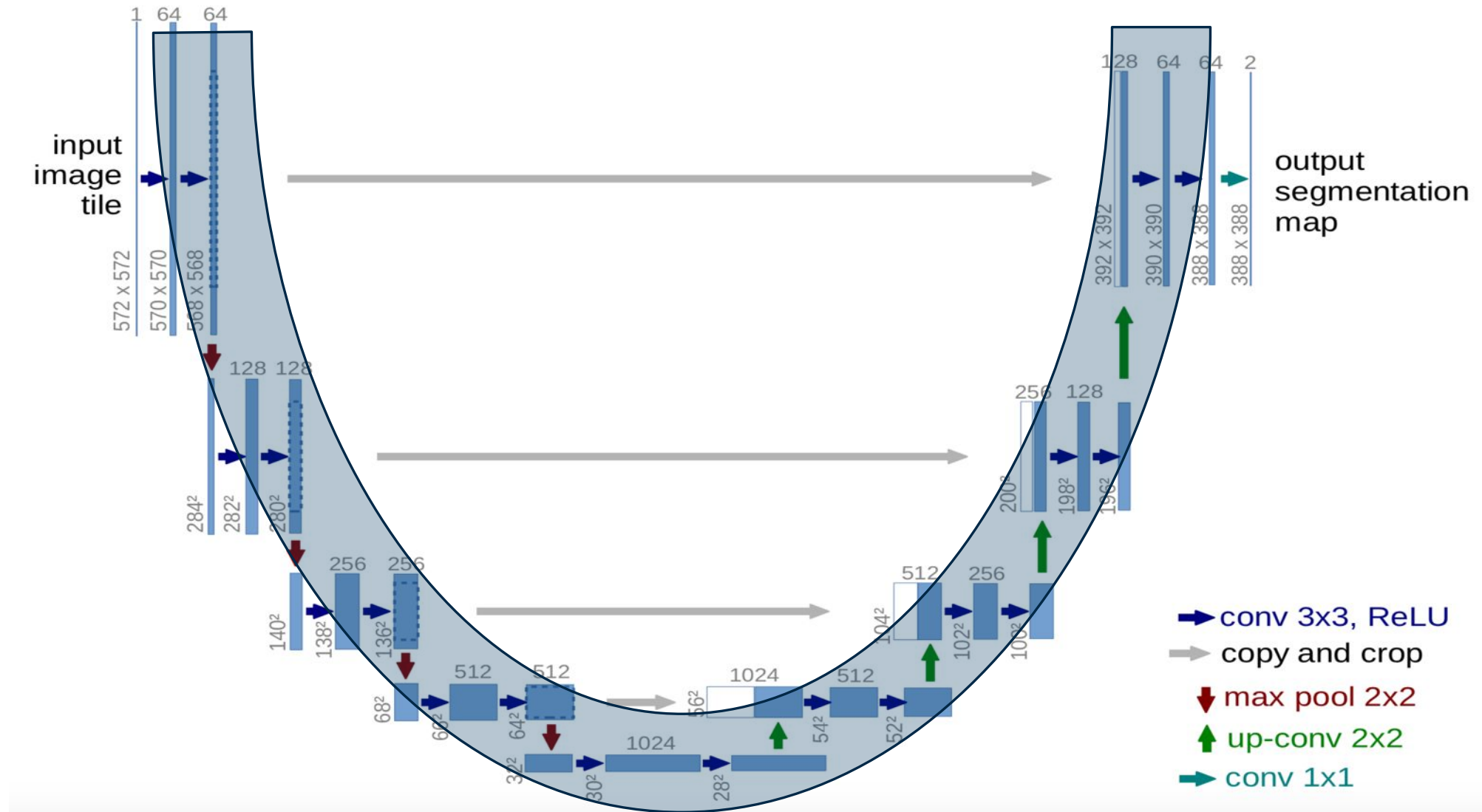
Task: segment neuronal structures from electron microscopy images



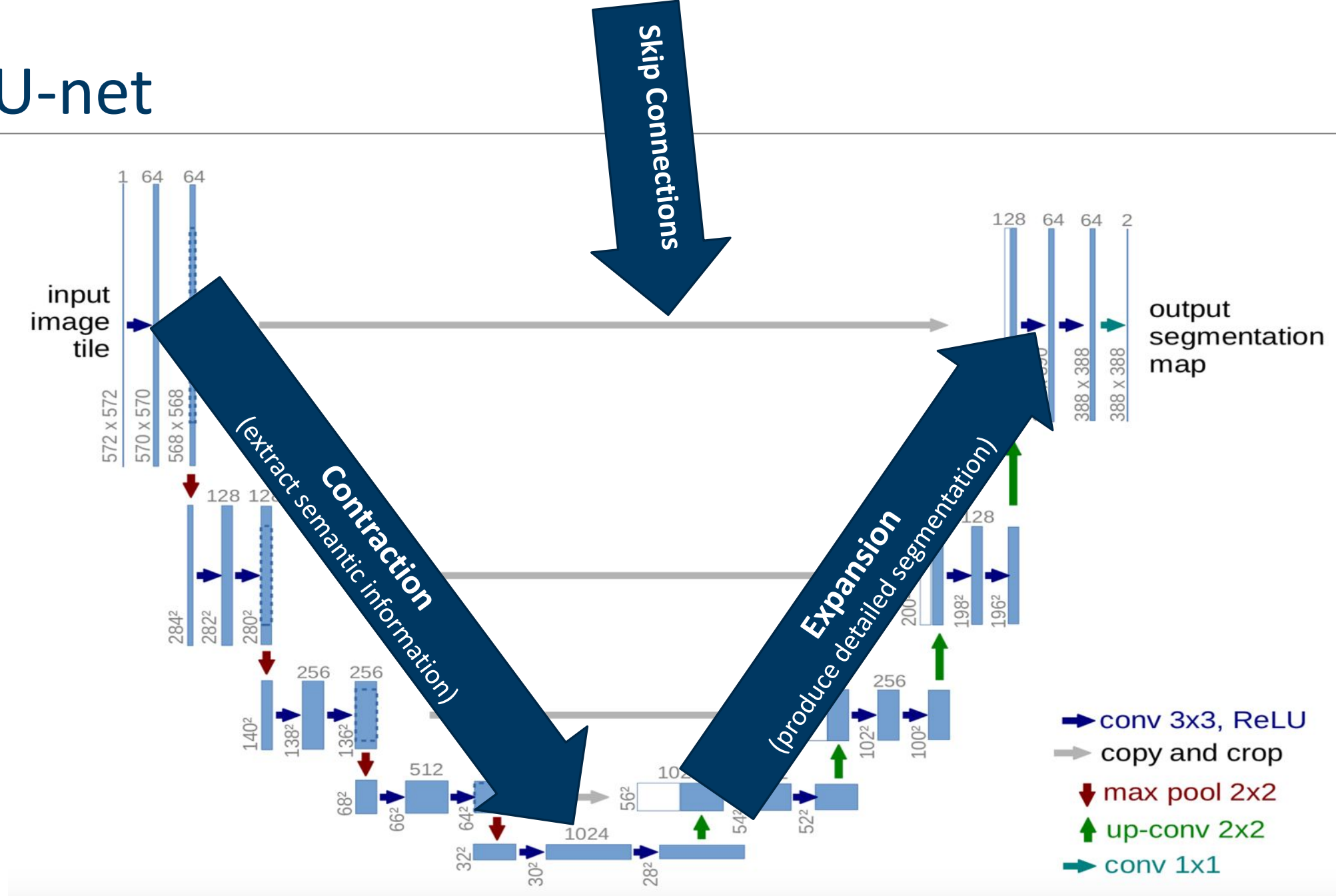
U-net



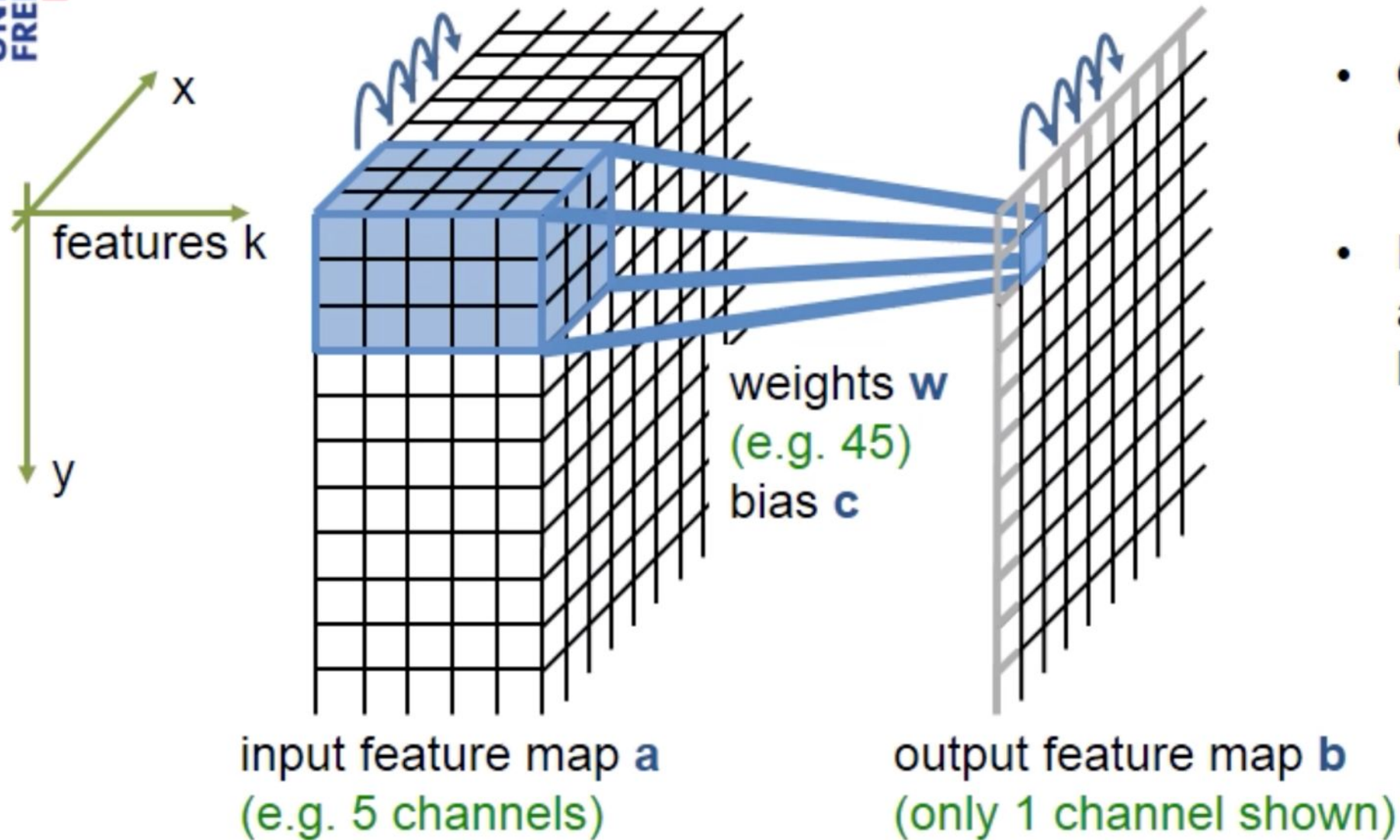
U-net



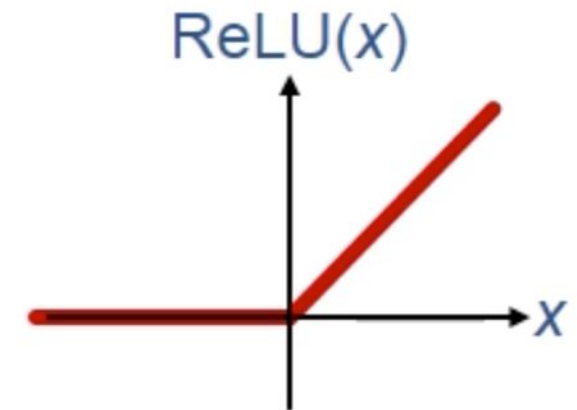
U-net



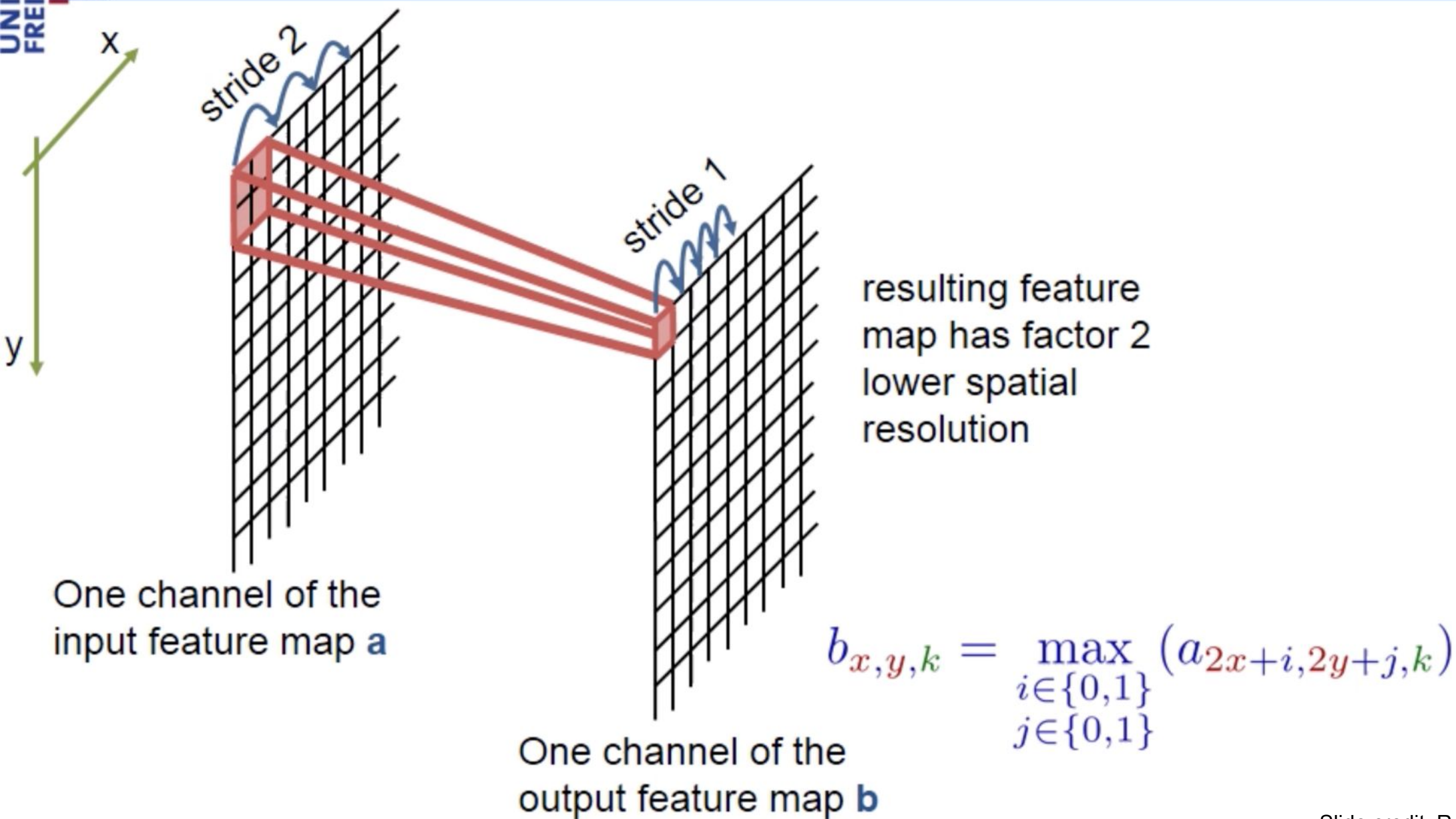
3x3 convolution + ReLU



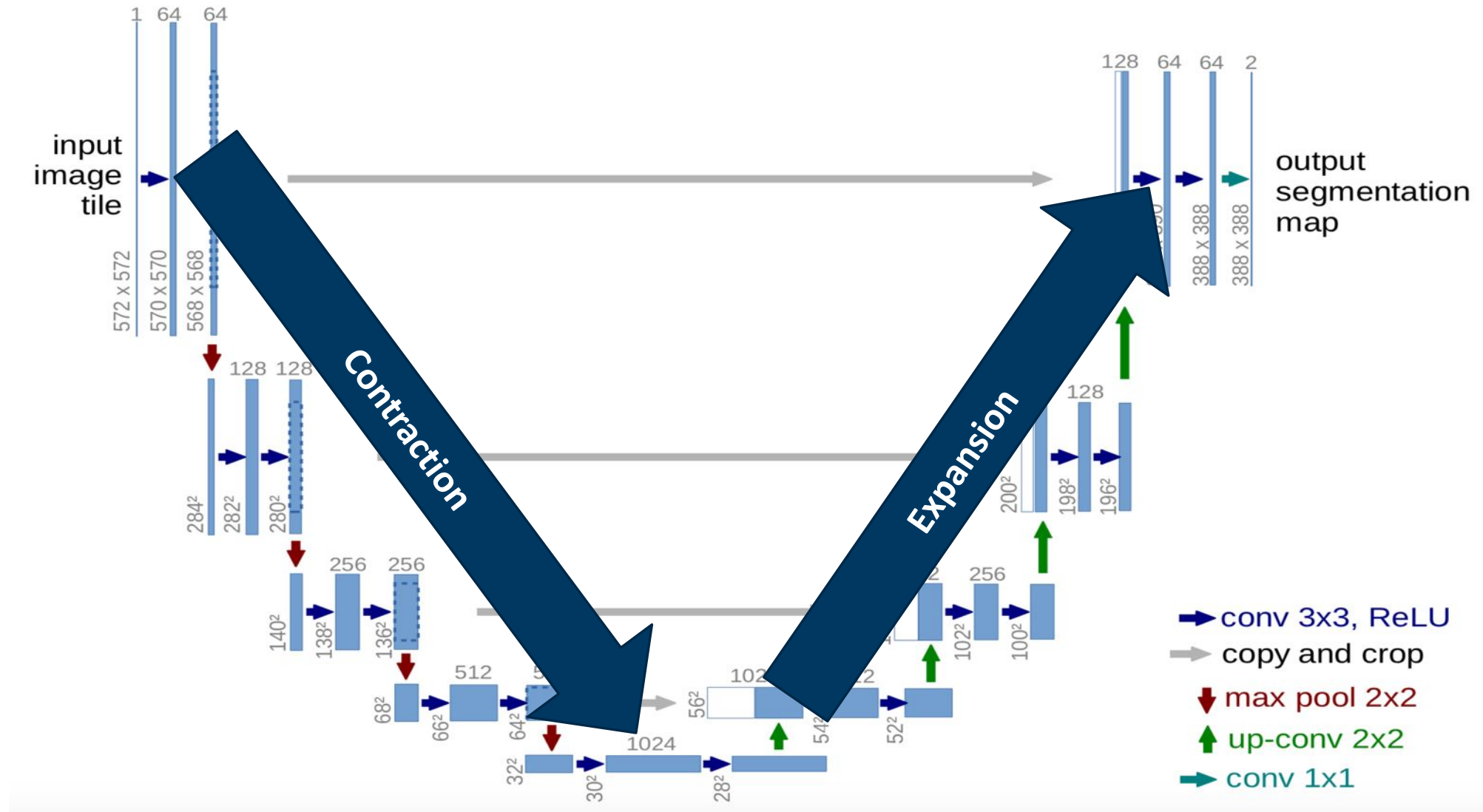
- Only valid part of convolution is used.
- For 3x3 convolutions a 1-pixel border is lost



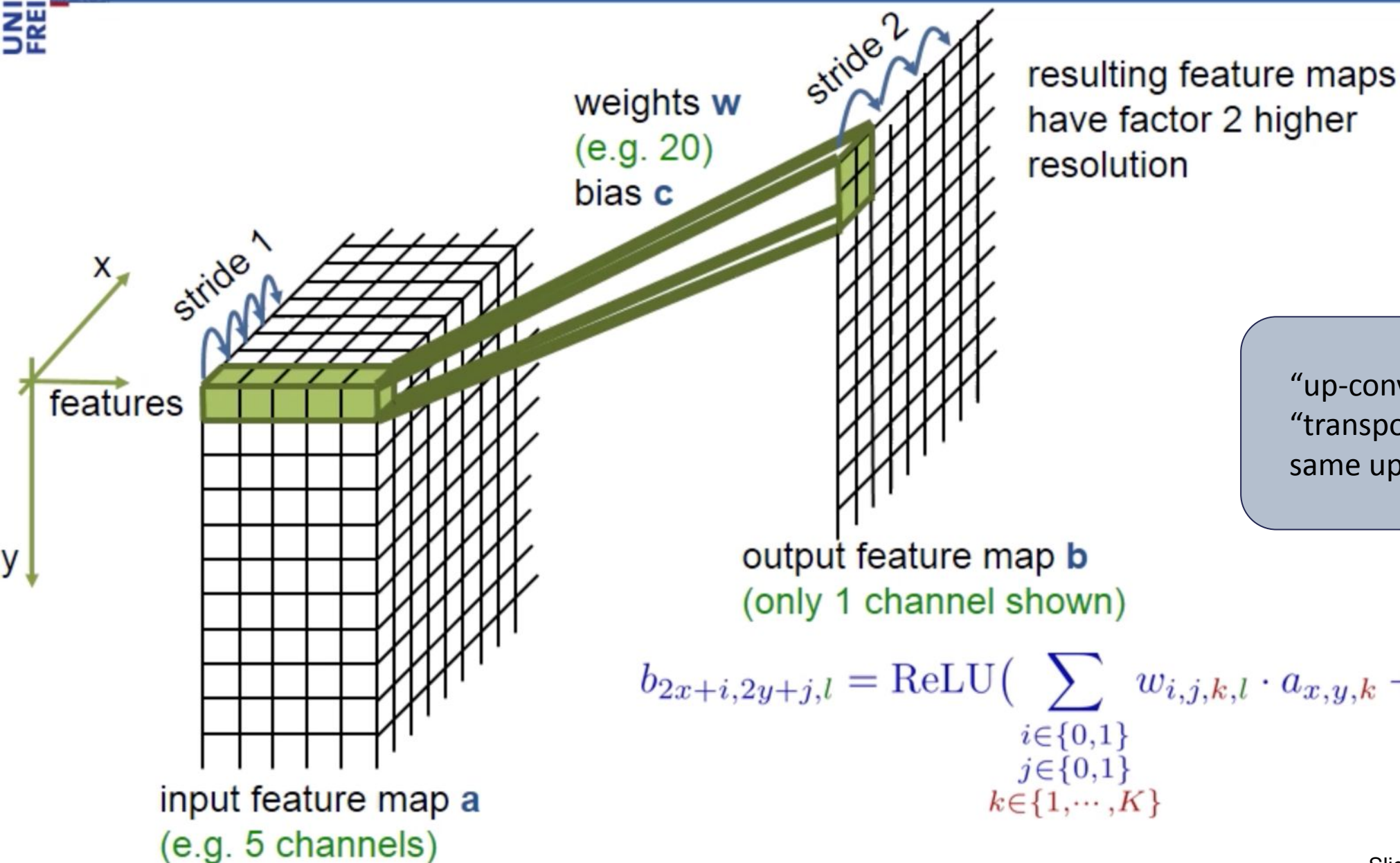
2x2 max-pooling



U-net



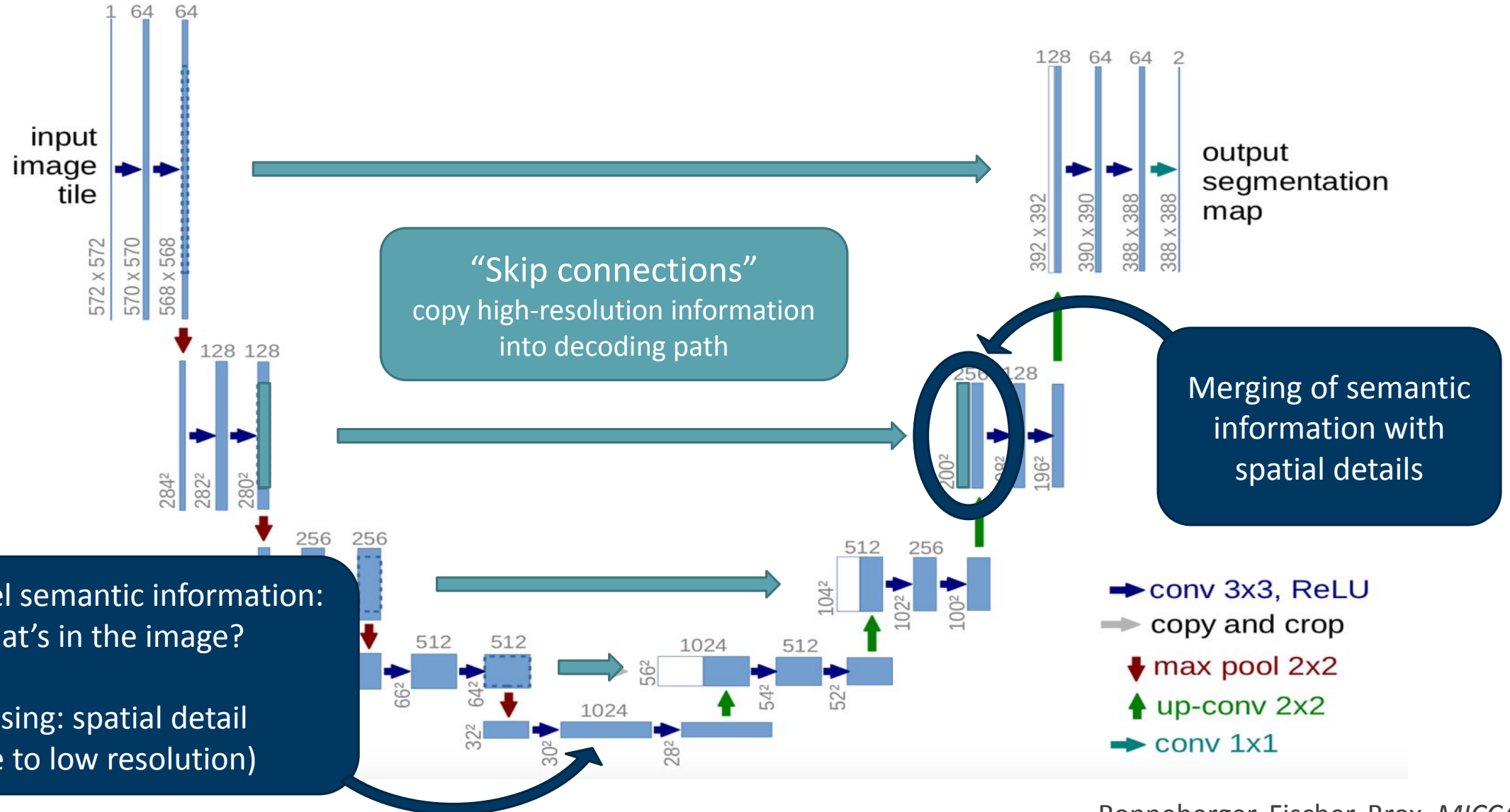
2x2 up-convolution



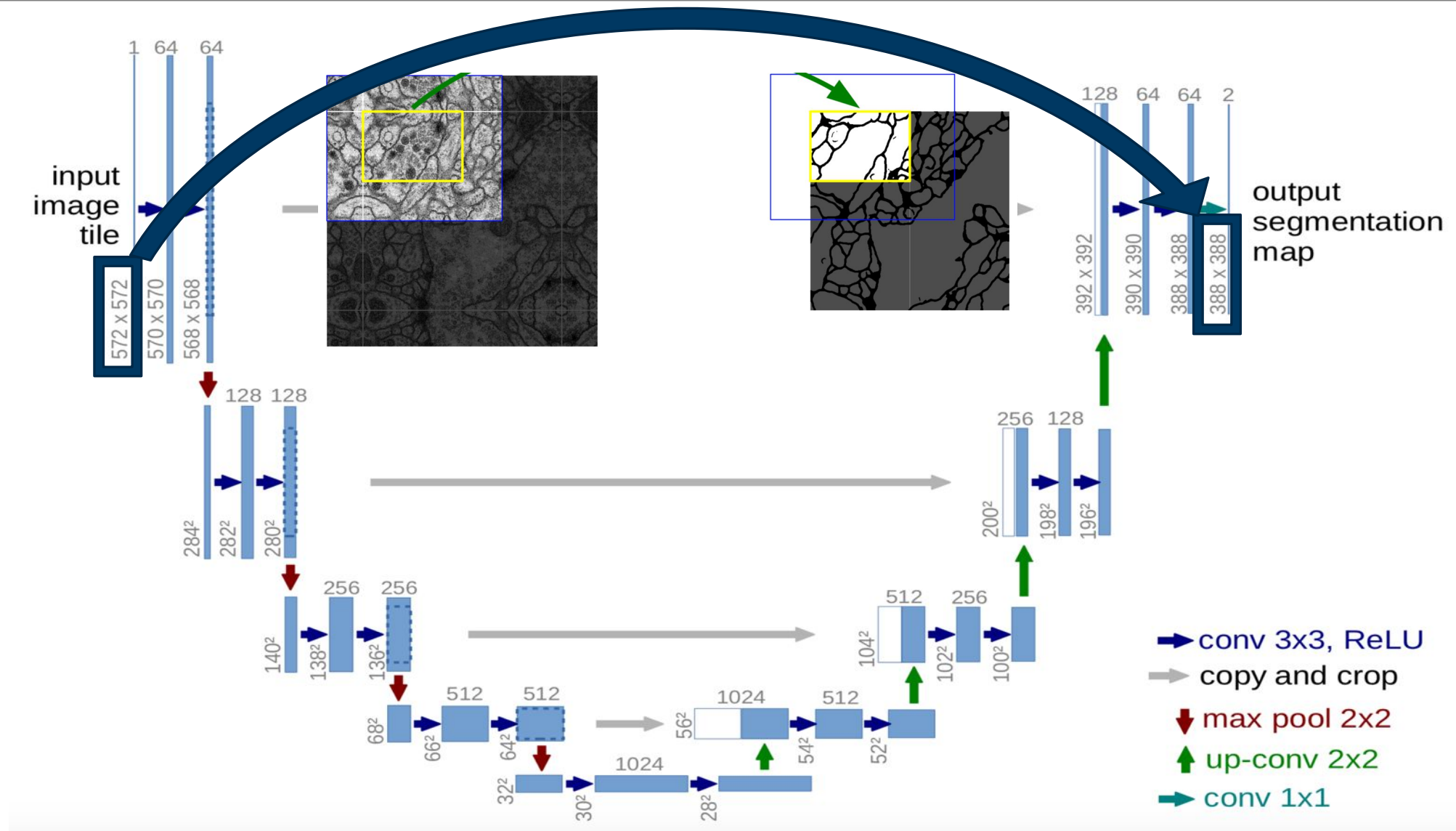
“up-convolution” =
“transposed convolution”
same upsampling as in FCN

$$b_{2x+i,2y+j,l} = \text{ReLU}\left(\sum_{\substack{i \in \{0,1\} \\ j \in \{0,1\} \\ k \in \{1,\dots,K\}}} w_{i,j,k,l} \cdot a_{x,y,k} + c_l\right)$$

Skip connections

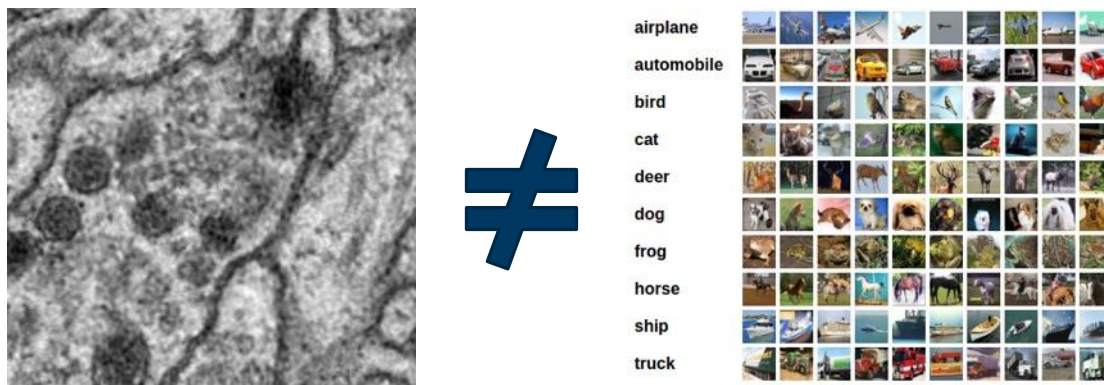


Spatial dimensions

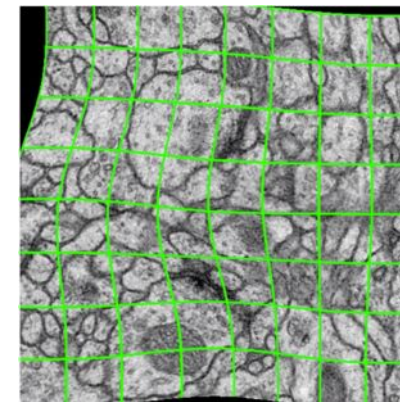


U-net: training

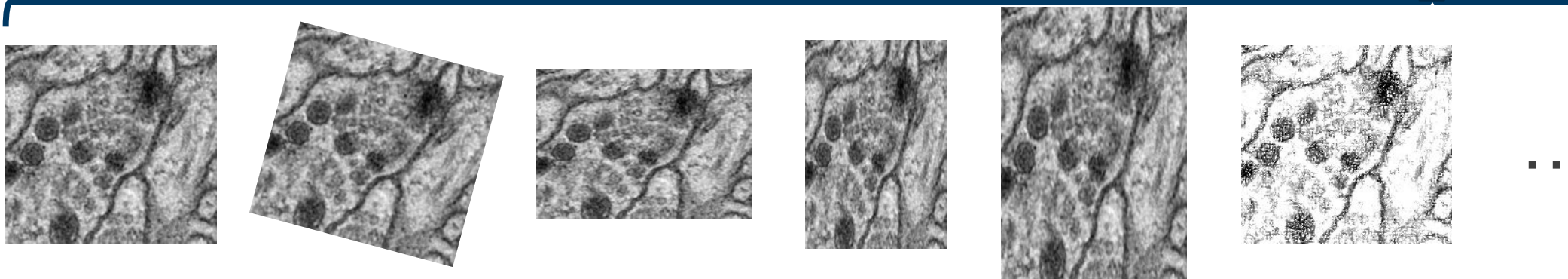
Often no pre-training on ImageNet



Elastic image deformations



Instead: heavy data augmentation

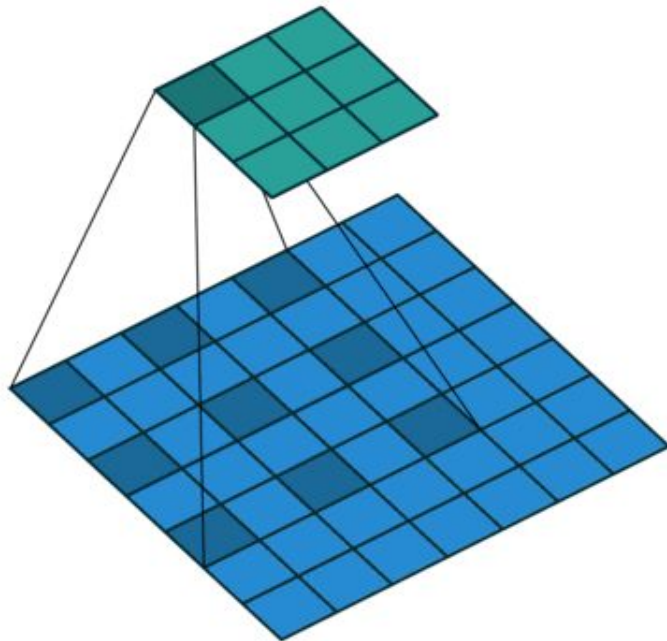
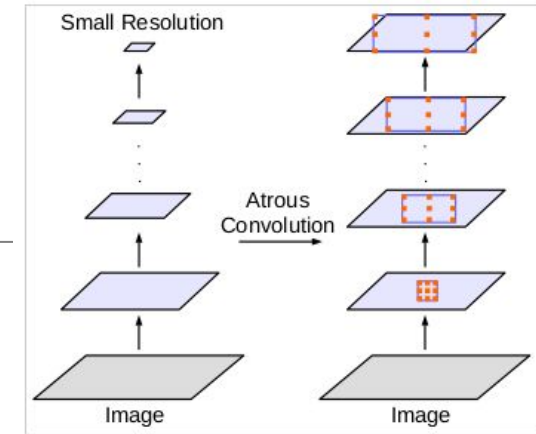


Architecture improvements

- [Holistically nested edge detection](#): multi-scale input image combination
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Deep Lab v1

Reminder: atrous (or dilated convolution):
dilate the filter by implicit zeros in between the kernel elements



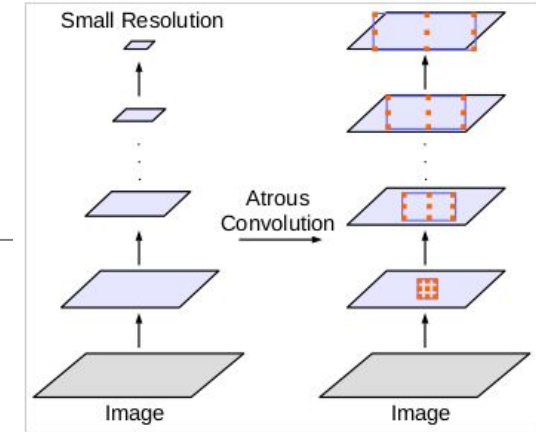
Notation: rate r determines dilation factor

$$y[i] = \sum_{k=1}^K x[i + r \cdot k]w[k]$$

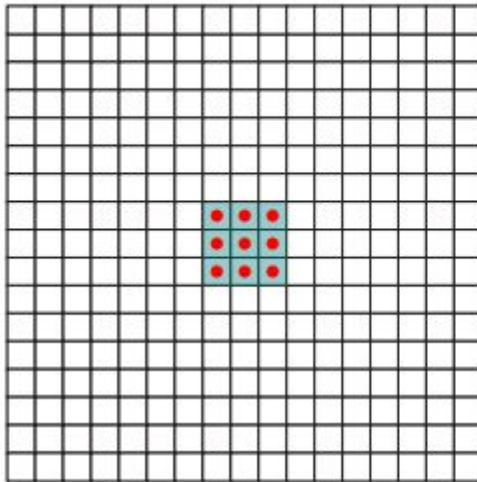
Deep Lab v1

Atrous convolutions enable larger receptive field without the loss of resolution (image size does not decrease)

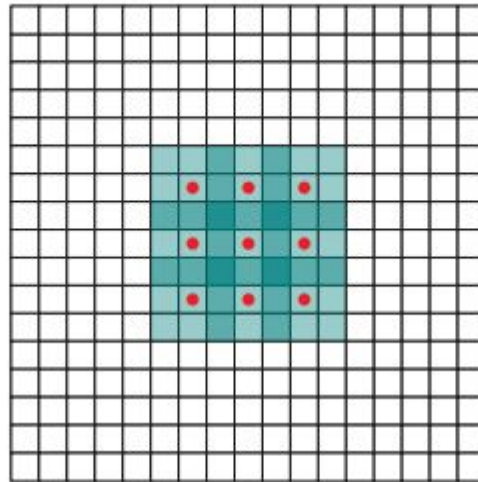
Example: 3x3 kernel, r = rate, RF = receptive field



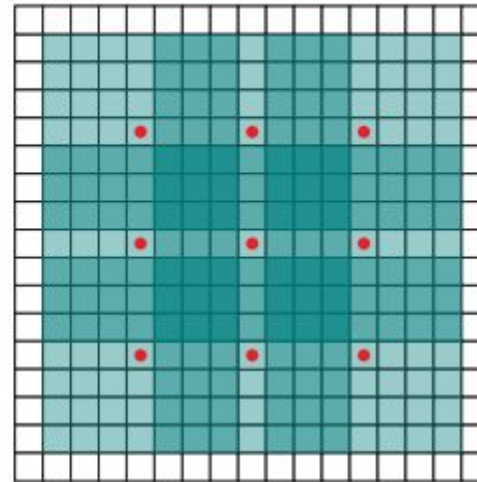
$r=1$, RF=3



$r=2$, RF=7

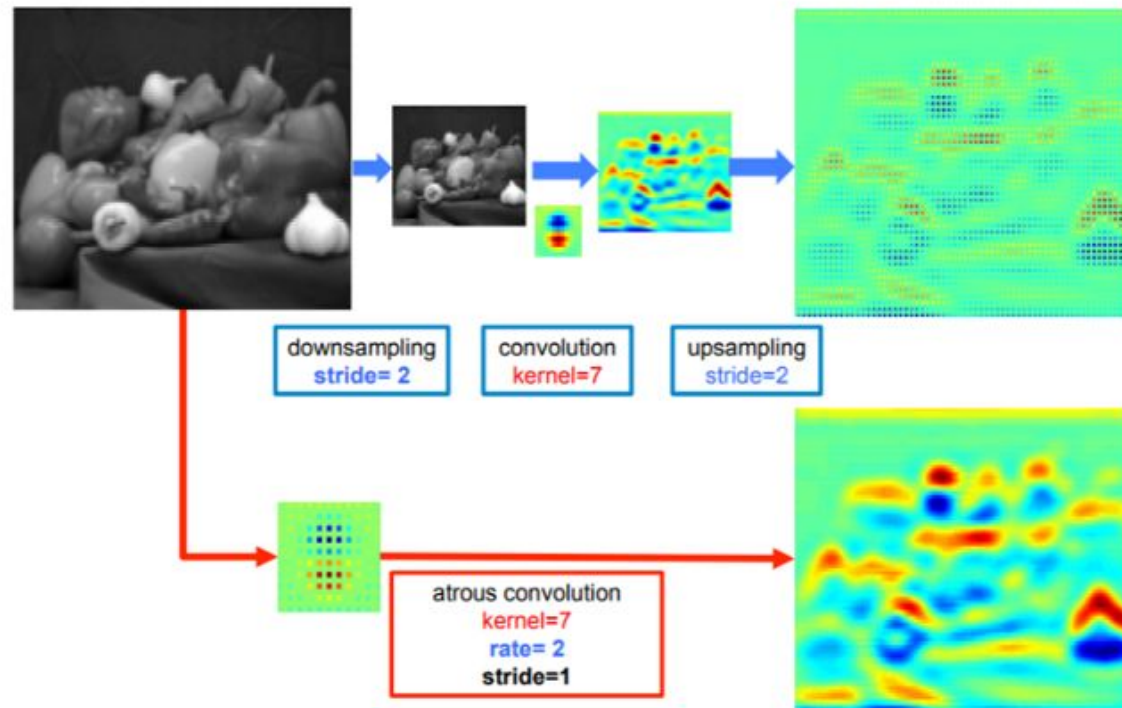
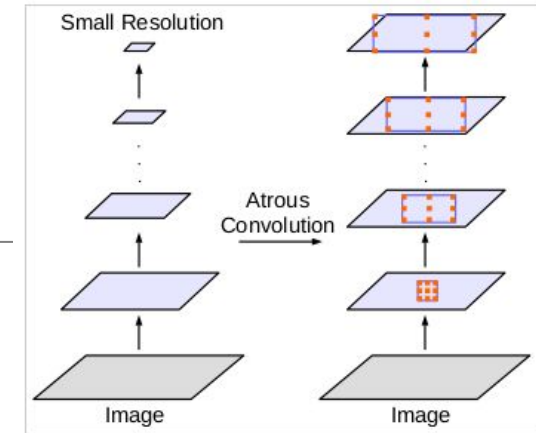


$r=4$, RF=15



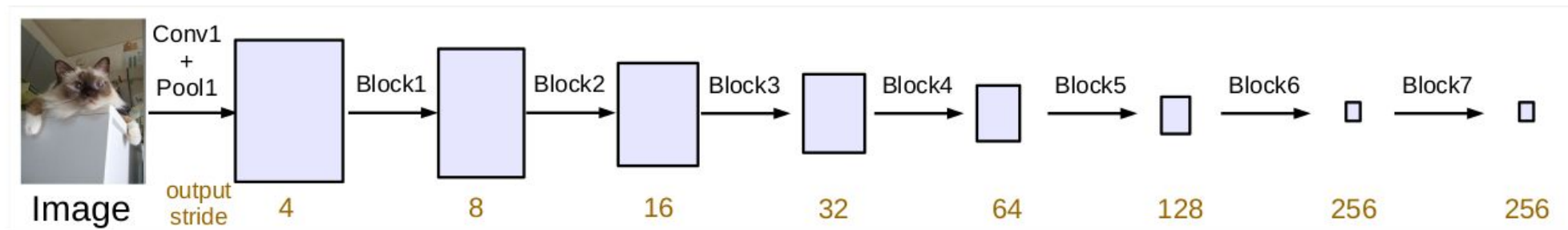
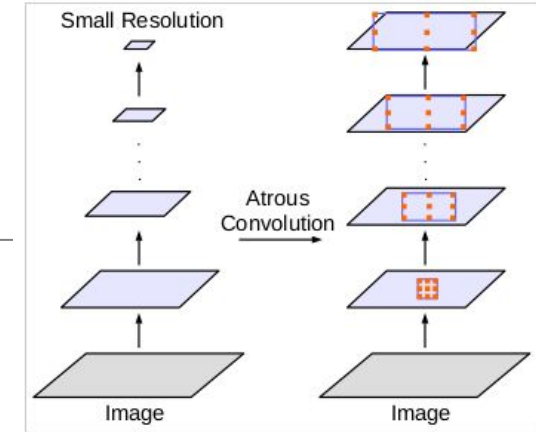
Deep Lab v1

Motivation: higher resolution feature maps compared to donwnsampling -> convolution -> upsampling



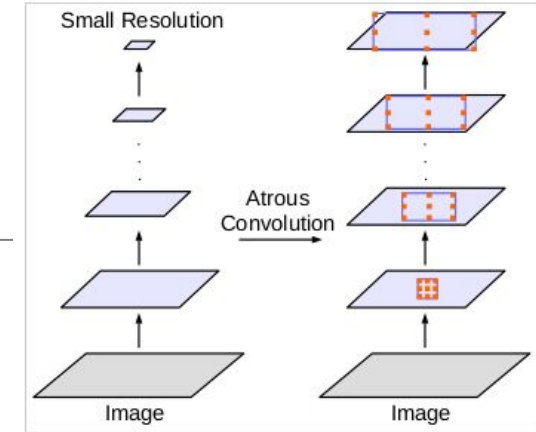
Deep Lab v1

Standard CNN: repeated strided convolution increases receptive field, but decreases resolution (image size)

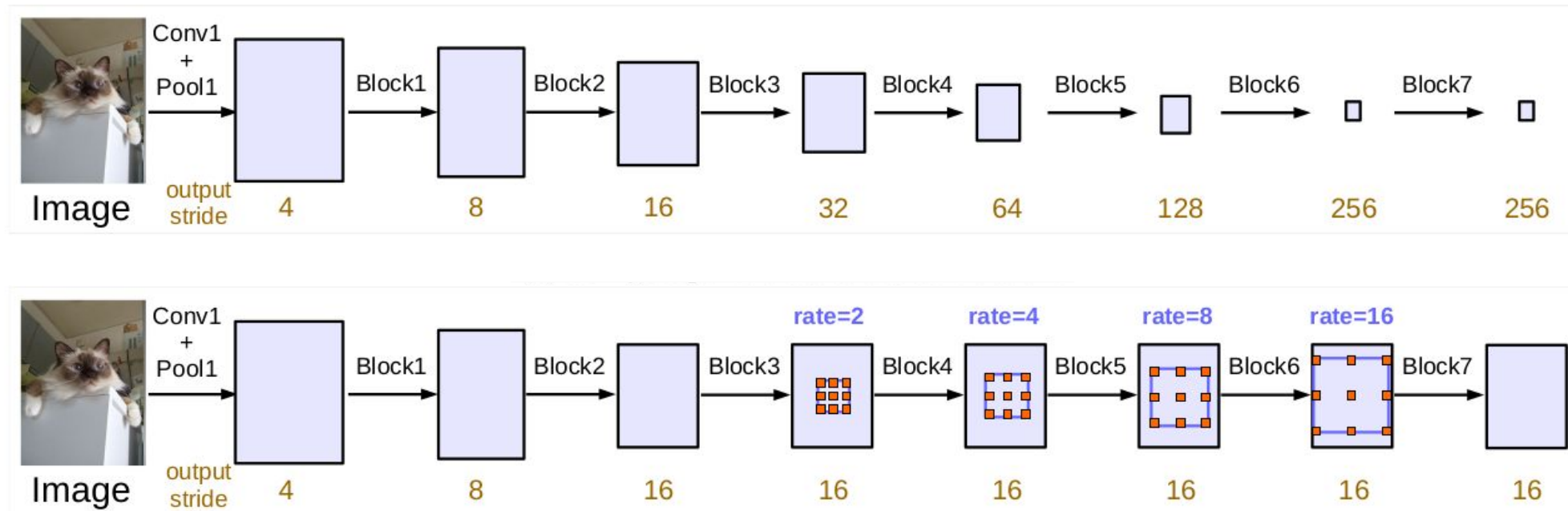


Deep Lab v1

Standard CNN: repeated strided convolution increases receptive field, but decreases resolution (image size)

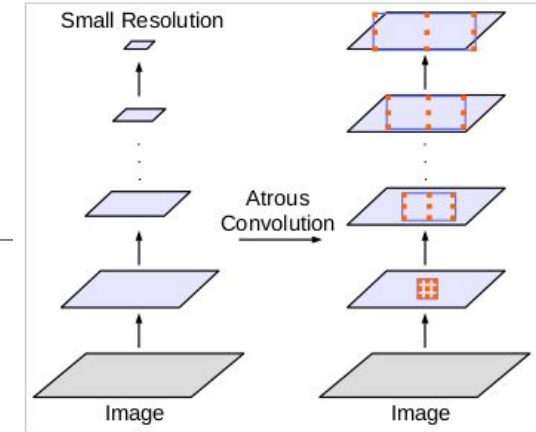


DeepLab v1: use atrous convolutions to avoid reduced resolution

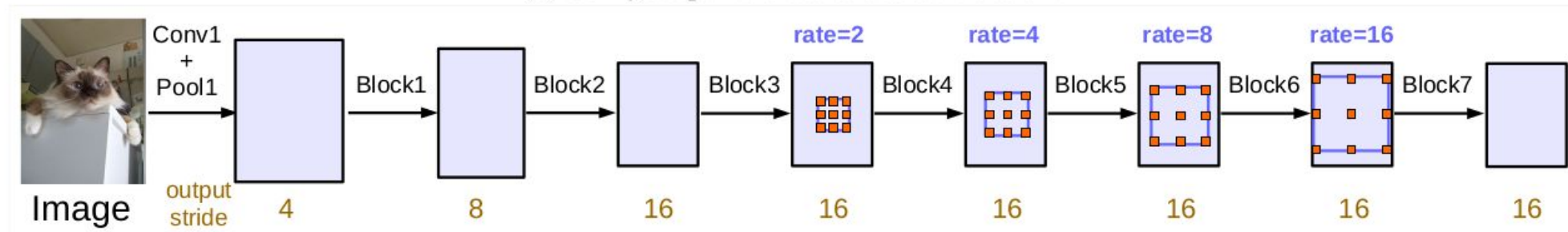


Deep Lab v1

Transfer Learning: backbone network is pre-trained on ImageNet



CNN for classification, pretrained on ImageNet. (VGG16 or ResNet101)



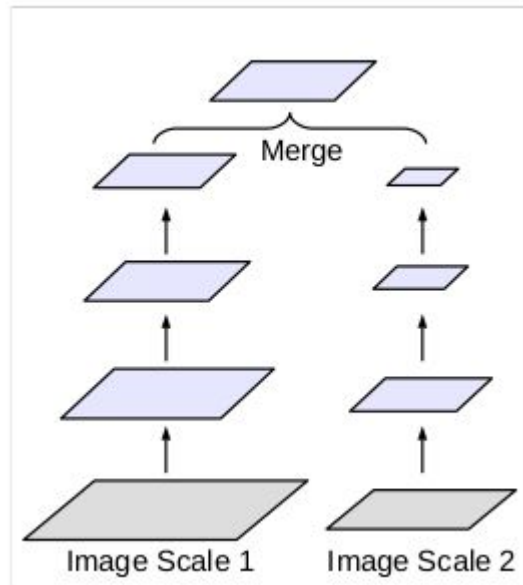
Take backbone as is

Replace strided conv / downsampling
with atrous conv

Learning Multi-scale Features

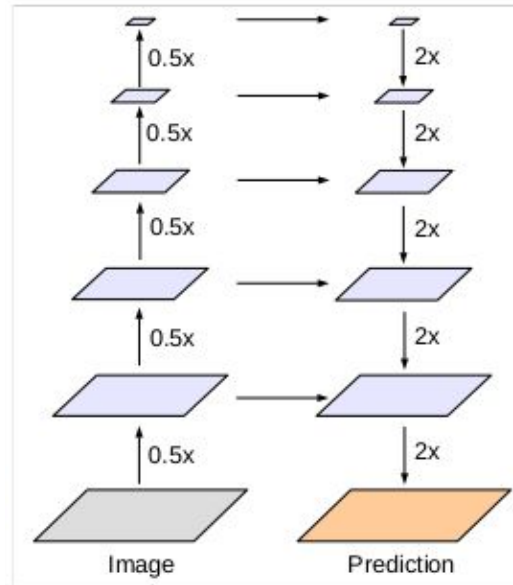
Goal: learn and combine feature representations at multiple scales

Image Pyramid



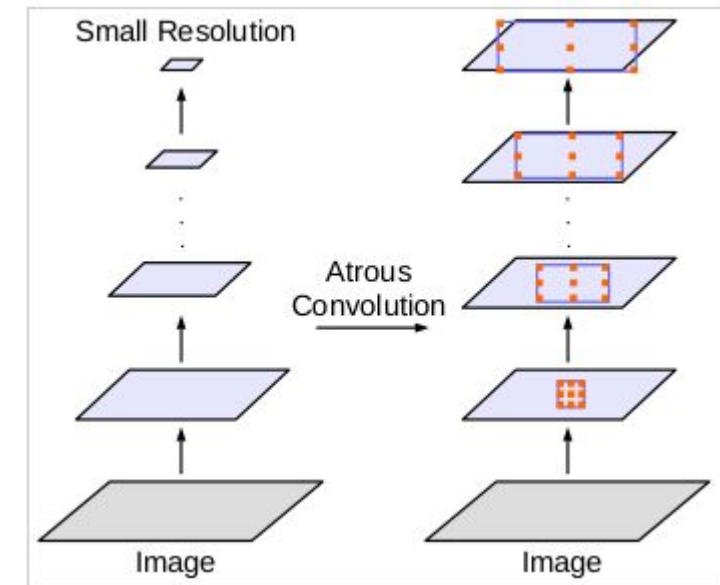
Holistically Nested Edge Detection

Encoder-Decoder



U-Net

Atrous Convolutions



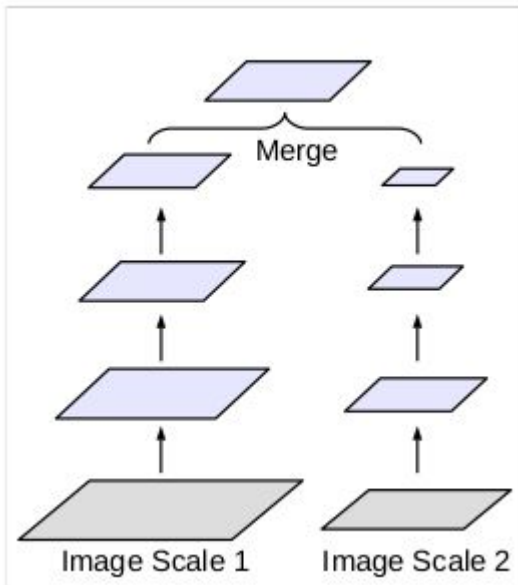
DeepLab v1

What's the best strategy?

Learning Multi-scale Features

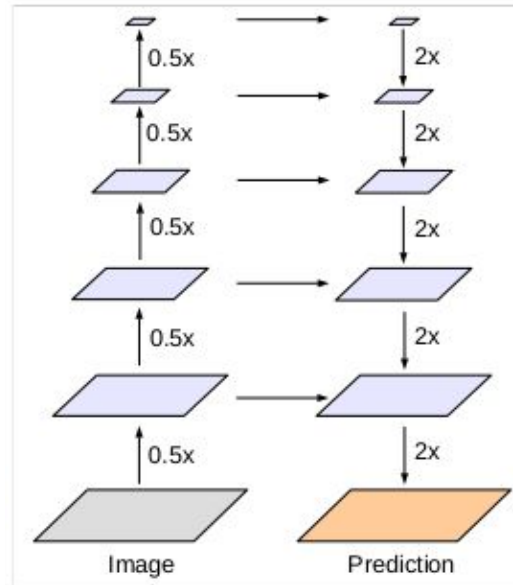
Goal: learn and combine feature representations at multiple scales

Image Pyramid



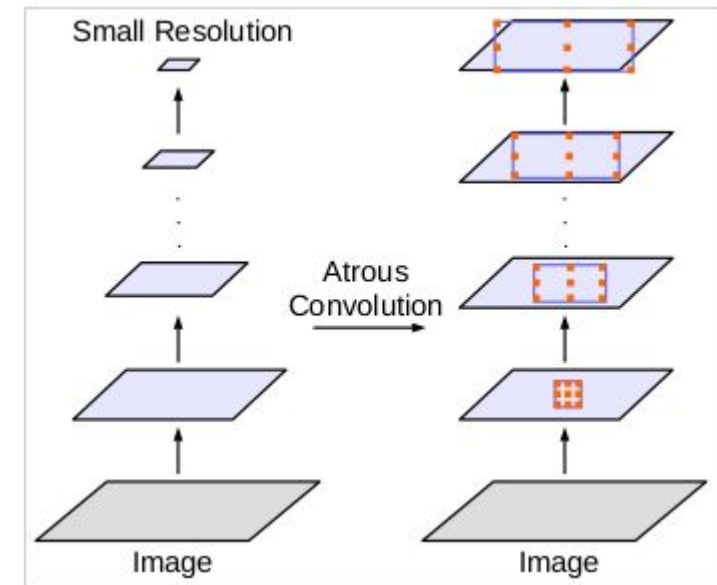
Holistically Nested Edge Detection

Encoder-Decoder



U-Net

Atrous Convolutions



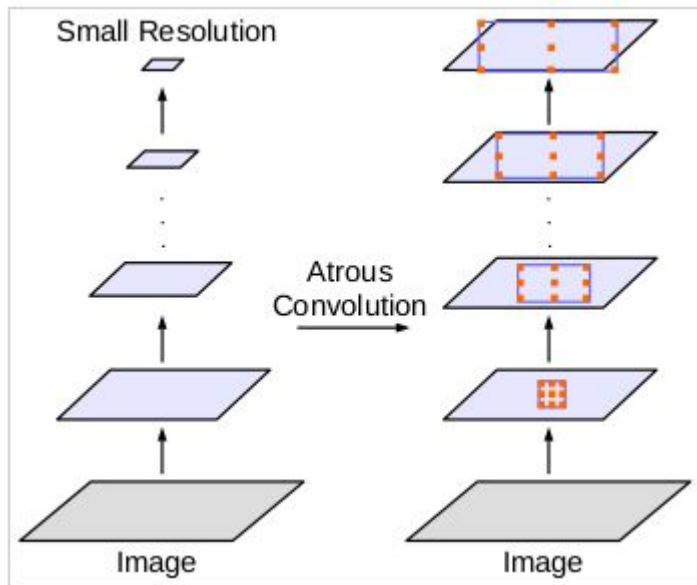
DeepLab v1

Combine strategies!

Learning Multi-scale Features

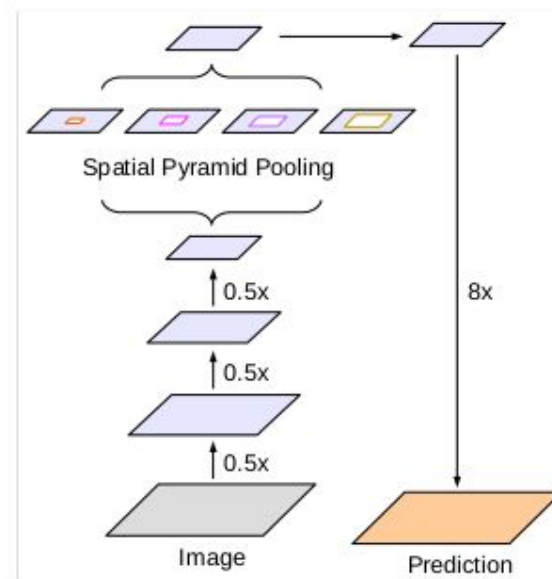
Goal: learn and combine feature representations at multiple scales

Atrous Convolutions



DeepLab v1

Spatial Pyramid Pooling (SPP)



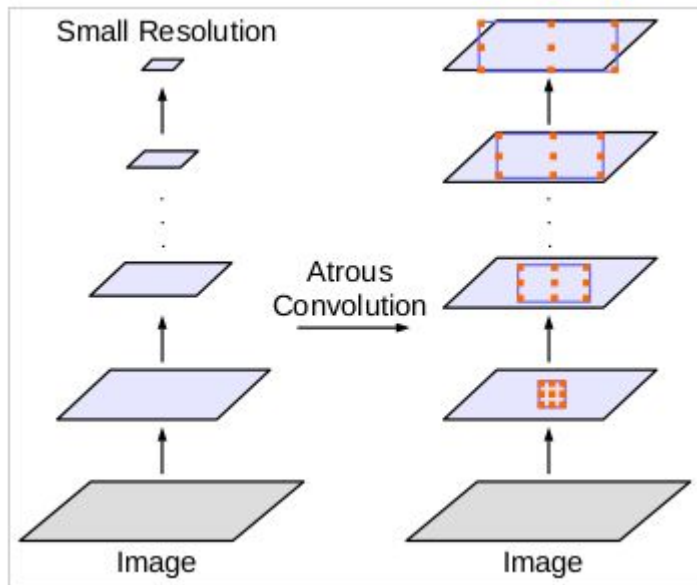
DeepLab v2

Combine strategies!

Learning Multi-scale Features

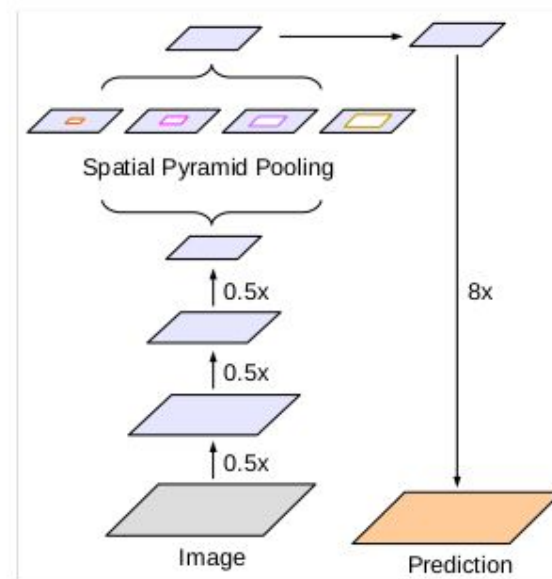
Goal: learn and combine feature representations at multiple scales

Atrous Convolutions



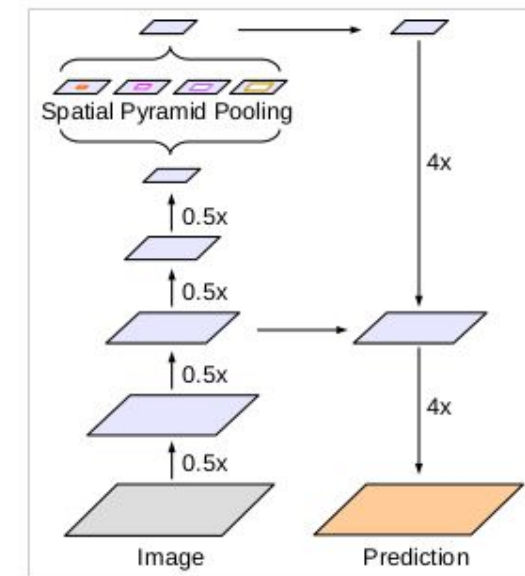
DeepLab v1

Spatial Pyramid Pooling (SPP)



DeepLab v2

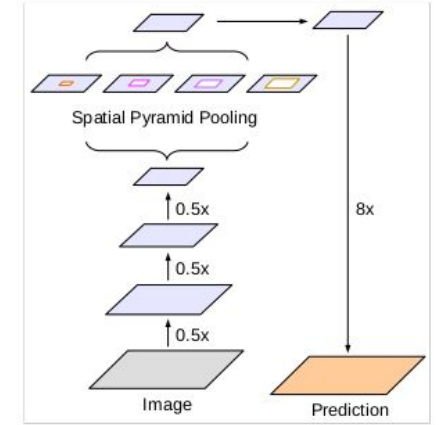
Encoder-Decoder with SPP



DeepLab v3

Deep Lab v2

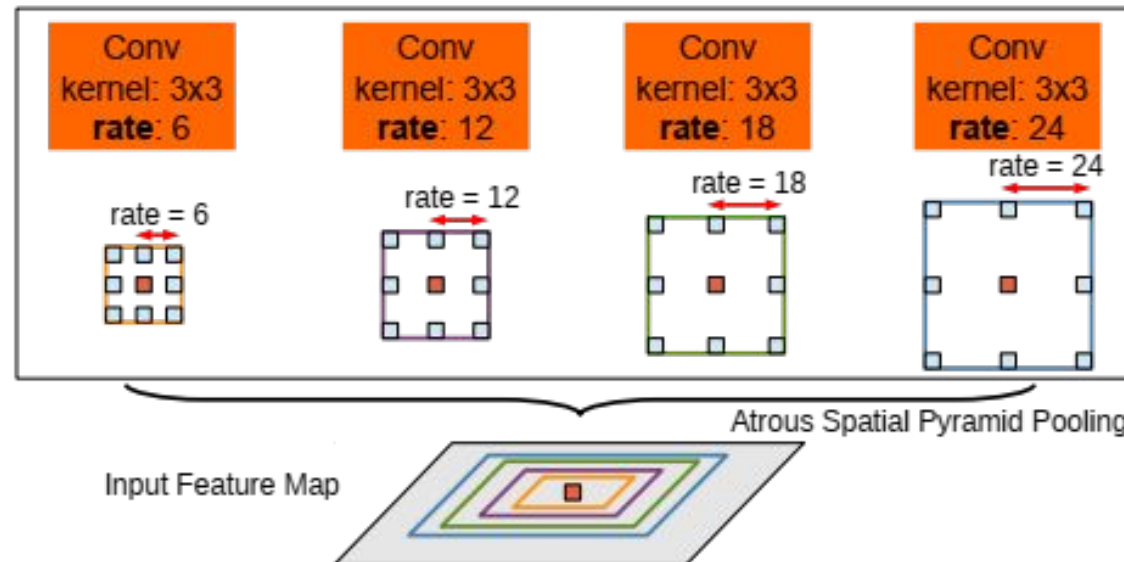
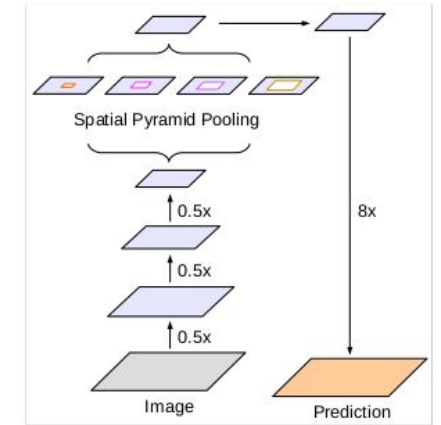
DeepLab v1: sequence of atrous convolutions, **but no multiscale**



Deep Lab v2

DeepLab v1: sequence of atrous convolutions, **but no multiscale**

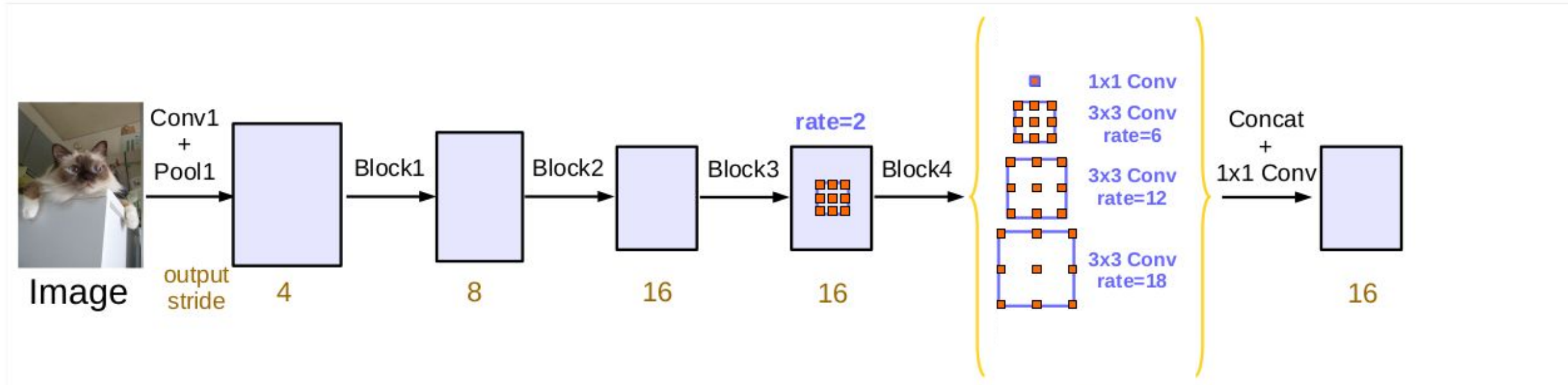
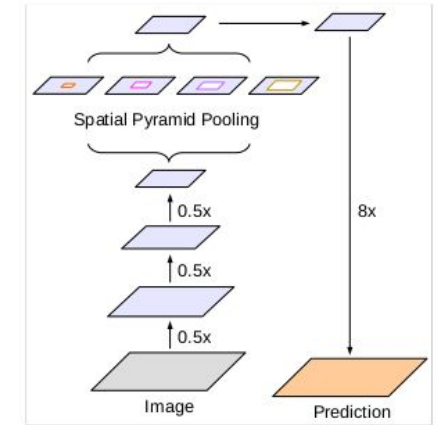
DeepLab v2: Atrous Spatial Pyramid Pooling (ASPP) to exploit multiple scales



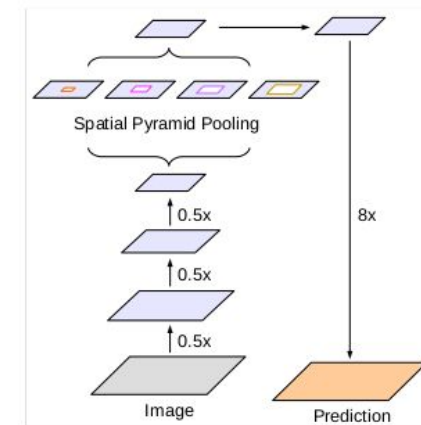
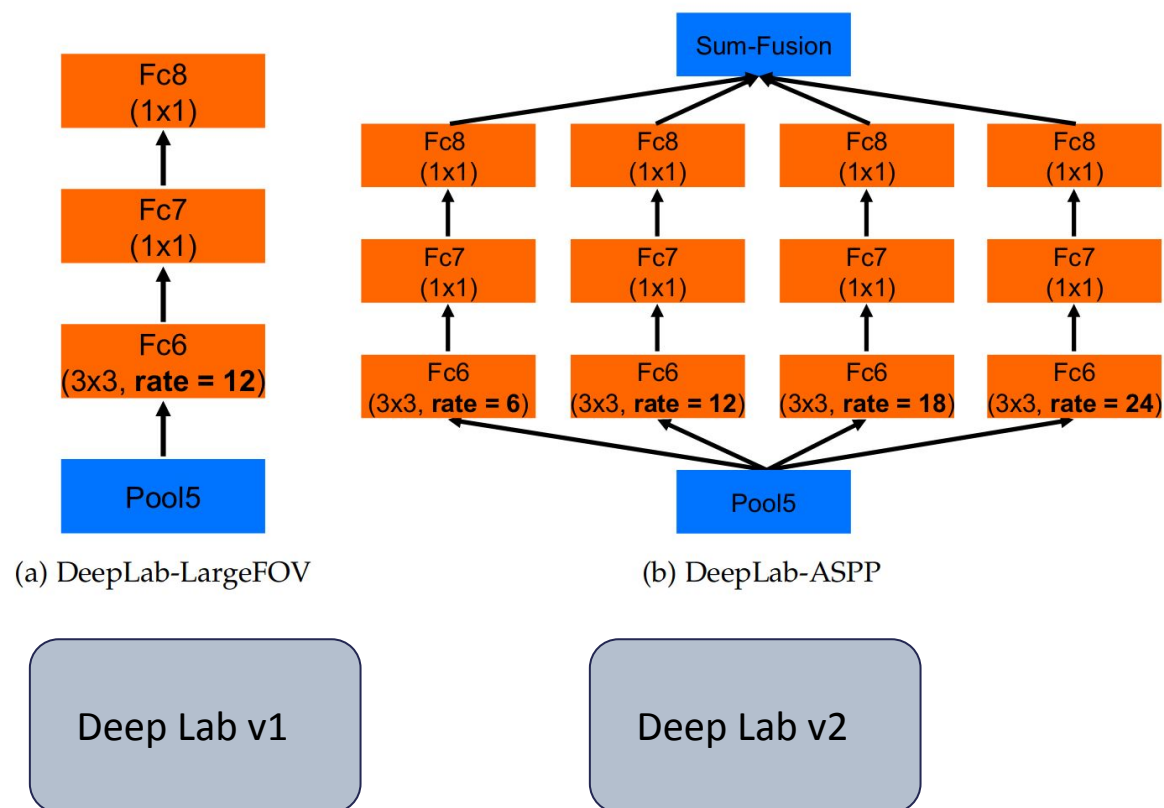
Deep Lab v2

DeepLab v1: sequence of atrous convolutions, **but no multiscale**

DeepLab v2: Atrous Spatial Pyramid Pooling (ASPP) to exploit multiple scales



Deep Lab v2



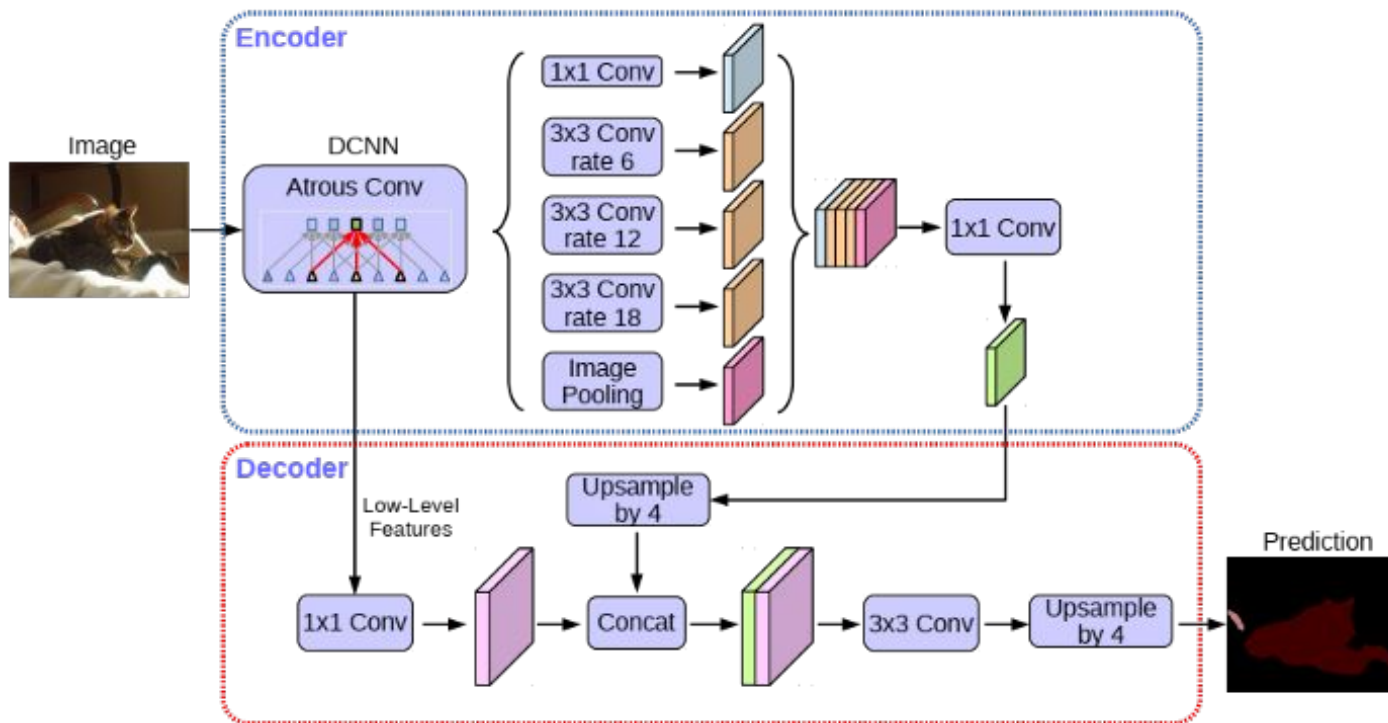
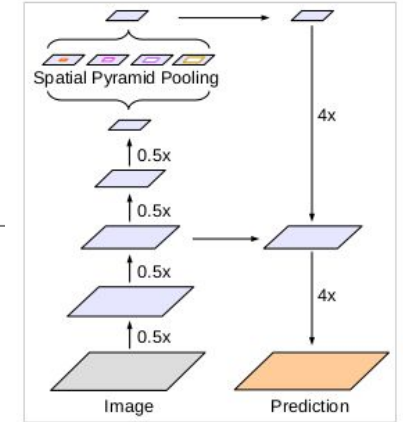
Method	mean IOU
LargeFOV	65.76
ASPP-S	66.98
ASPP-L	68.96

Results on Pascal VOC

Deep Lab v3

DeepLab v2: ASPP for parallel multi-scale, **but coarse upsampling**

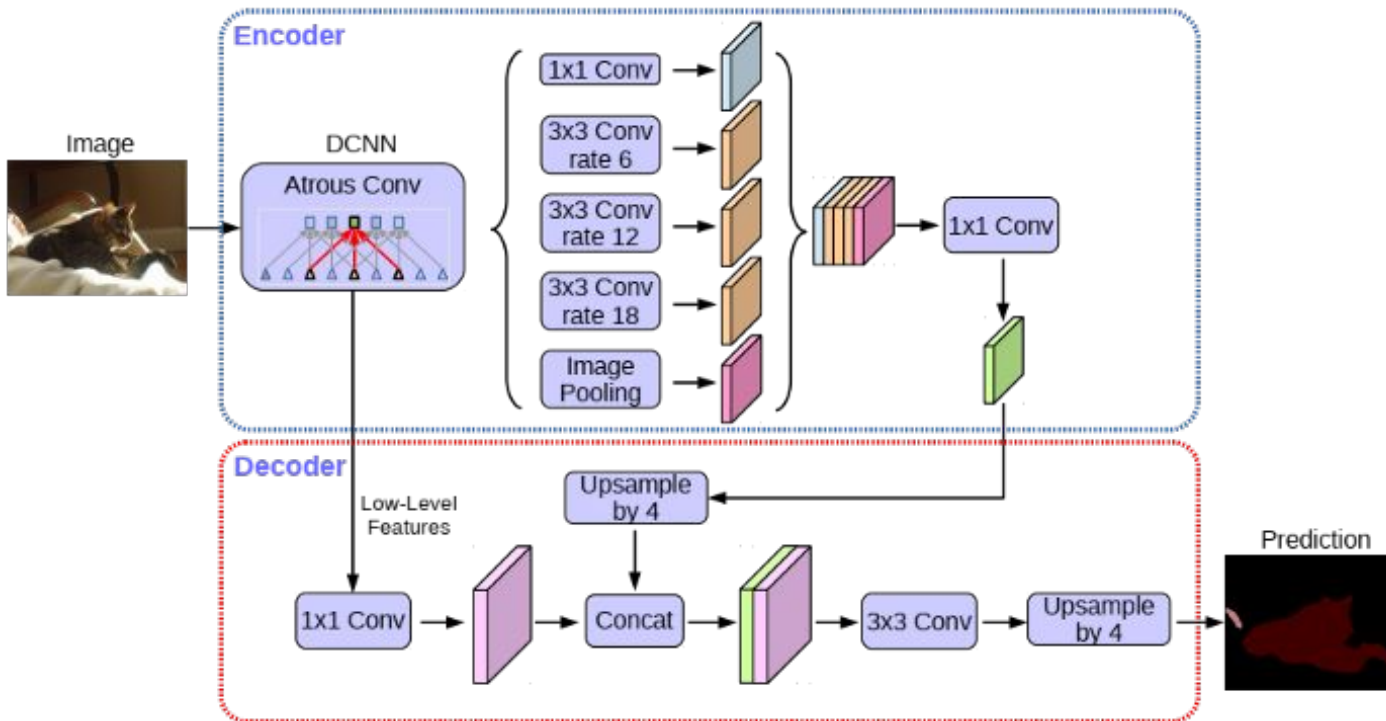
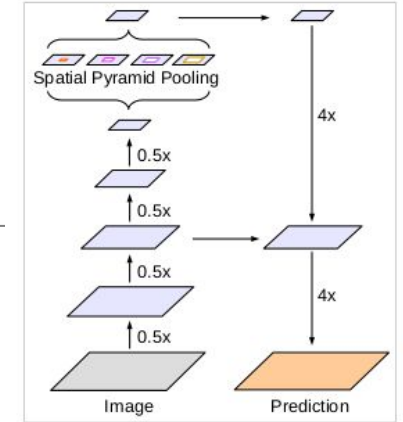
DeepLab v3: combine ASPP encoder with decoder



Deep Lab v3

DeepLab v2: ASPP for parallel multi-scale, **but coarse upsampling**

DeepLab v3: combine ASPP encoder with decoder



DeepLab v3: uses ResNet in Encoder

DeepLab v3+:
uses Encoder and Decoder with
depthwise separable convolutions

Good overview article:

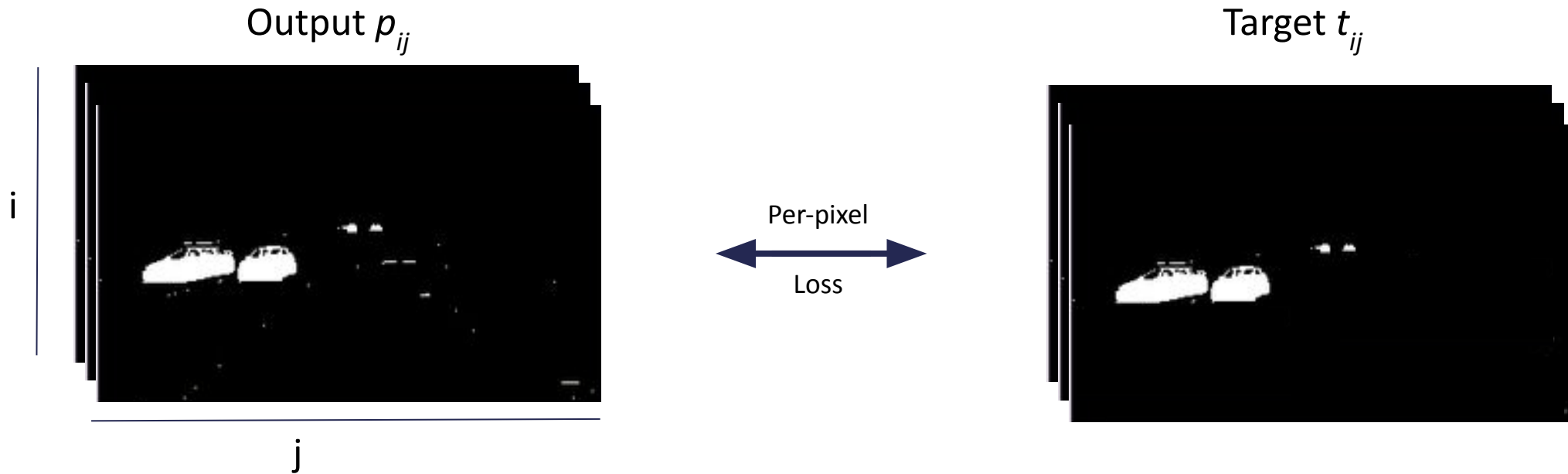
<https://towardsdatascience.com/the-evolution-of-deeplab-for-semantic-segmentation-95082b025571>

Semantic Segmentation

Loss Functions

Loss functions revisited

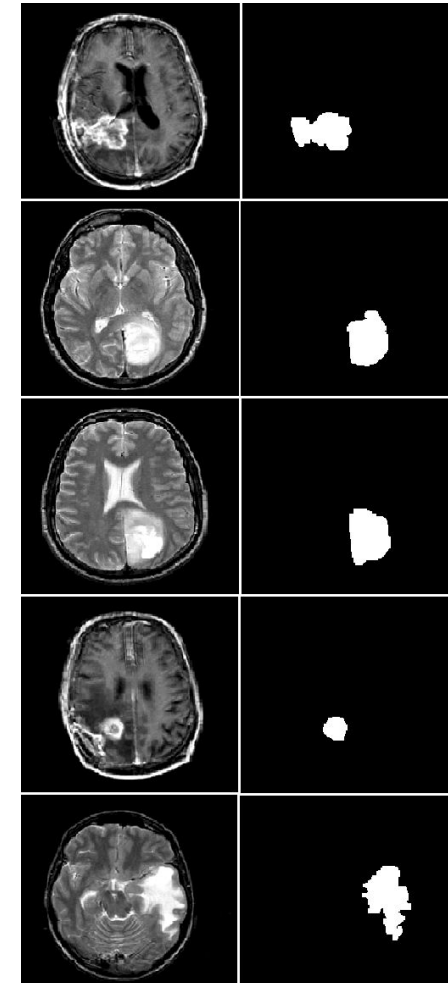
Apply (binary) cross entropy pixel-wise: $-\frac{1}{N} \sum_{ij} \sum_k \log p_{ij}^k t_{ij}^k$



Loss functions revisited

Problem: segmentation task is often imbalanced:

e.g. tumor segmentation: small foreground class



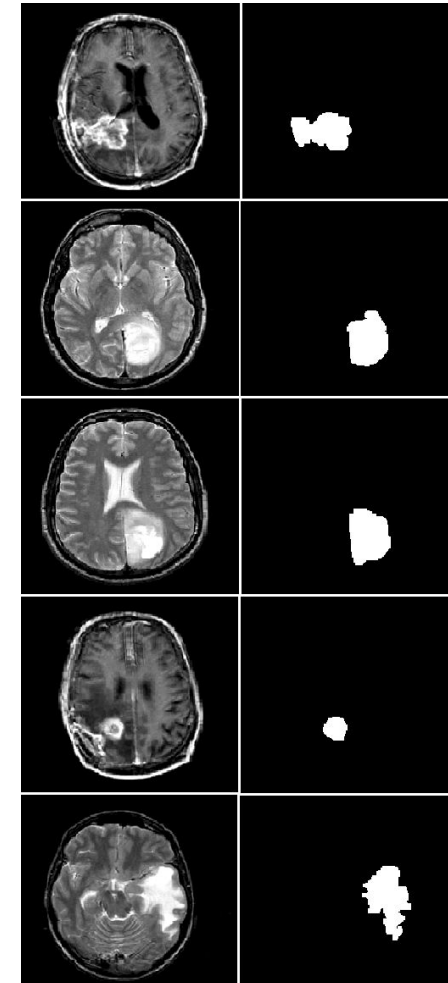
Loss functions revisited

Problem: segmentation task is often imbalanced:

e.g. tumor segmentation: small foreground class

(Binary) Cross Entropy:

predicting only background will give low loss



Weighted Cross Entropy

For simplicity: write sum over pixels ij (2d) as flat sum over n

Weighted Cross Entropy:

- class weight r inverse to class frequency
- (optional) spatial weight w

$$\text{WCE} = -\frac{1}{N} \sum_n^N \sum_k^K w_n r_k \log p_{n,k} t_{n,k}$$

Weighted Cross Entropy

For simplicity: write sum over pixels ij (2d) as flat sum over n

Weighted Cross Entropy:

- class weight r inverse to class frequency
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$$\text{WCE} = -\frac{1}{N} \sum_n \sum_k^K w_n r_k \log p_{n,k} t_{n,k}$$

r_1 : weight for foreground class

Weighted Binary Cross Entropy:

$$\text{WBCE} = -\frac{1}{N} \sum_n w_n (r_1 \log p_n + (1 - r_1) \log(1 - p_n))$$

Weighted Cross Entropy

Weighted Cross Entropy:

- class weight r inverse to class frequency
- (optional) spatial weight w

$$\text{WCE} = -\frac{1}{N} \sum_n \sum_k^K w_n r_k \log p_{n,k} t_{n,k}$$

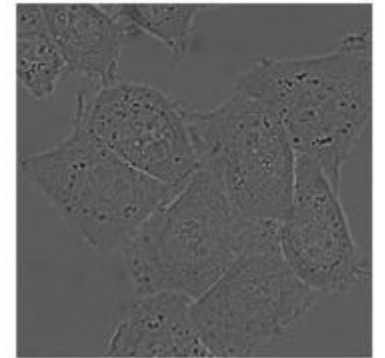
Weighted Binary Cross Entropy:

$$\text{WBCE} = -\frac{1}{N} \sum_n w_n (r_1 \log p_n + (1 - r_1) \log(1 - p_n))$$

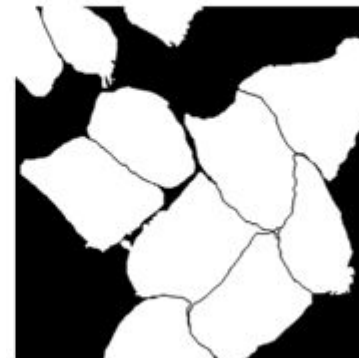
Example: Cell Segmentation
(Foreground/Background)

- class weight:
inverse class frequency
- spatial weight:
distance to cell

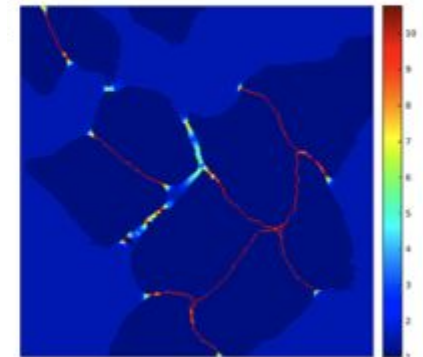
Input



Segmentation map



Pixel-wise loss weights



Dice Loss

Dice Loss: loss based on overlap of prediction p and target t

$$\text{DICE} = 1 - \frac{2 \sum_n^N t_n p_n}{\sum_n^N t_n^2 + \sum_n^N p_n^2}$$

Binary setting (same as BCE):

- t : binary target
(0: background, 1: foreground)
- p : foreground probabilities

Dice Loss

Dice Loss: loss based on overlap of prediction p and target t

$$\text{DICE} = 1 - \frac{2 \sum_n^N t_n p_n}{\sum_n^N t_n^2 + \sum_n^N p_n^2}$$

Dice measures overlap. Range:
0 (no overlap) to 1 (full overlap).
Use “1 - ” to minimize.

Binary setting (same as BCE):

- t : binary target
(0: background, 1: foreground)
- p : foreground probabilities

Dice Loss

Dice Loss: loss based on overlap of prediction p and target t

$$\text{DICE} = 1 - \frac{2 \sum_n^N t_n p_n}{\sum_n^N t_n^2 + \sum_n^N p_n^2}$$

Dice is “self normalizing”:

- small foreground class -> small denominator -> larger value
- large foreground class -> large denominator -> small value

Binary setting (same as BCE):

- t : binary target
(0: background, 1: foreground)
- p : foreground probabilities

Dice Loss

Dice Loss: loss based on overlap of prediction p and target t

$$\text{DICE} = 1 - \frac{2 \sum_n^N t_n p_n}{\sum_n^N t_n^2 + \sum_n^N p_n^2}$$

Binary setting (same as BCE):

- t : binary target
(0: background, 1: foreground)
- p : foreground probabilities

But: not well defined for empty samples (no foreground).
Combine with Binary Cross Entropy

$$\mathcal{L} = \alpha \mathcal{L}_{BCE} + \beta \mathcal{L}_{Dice}$$

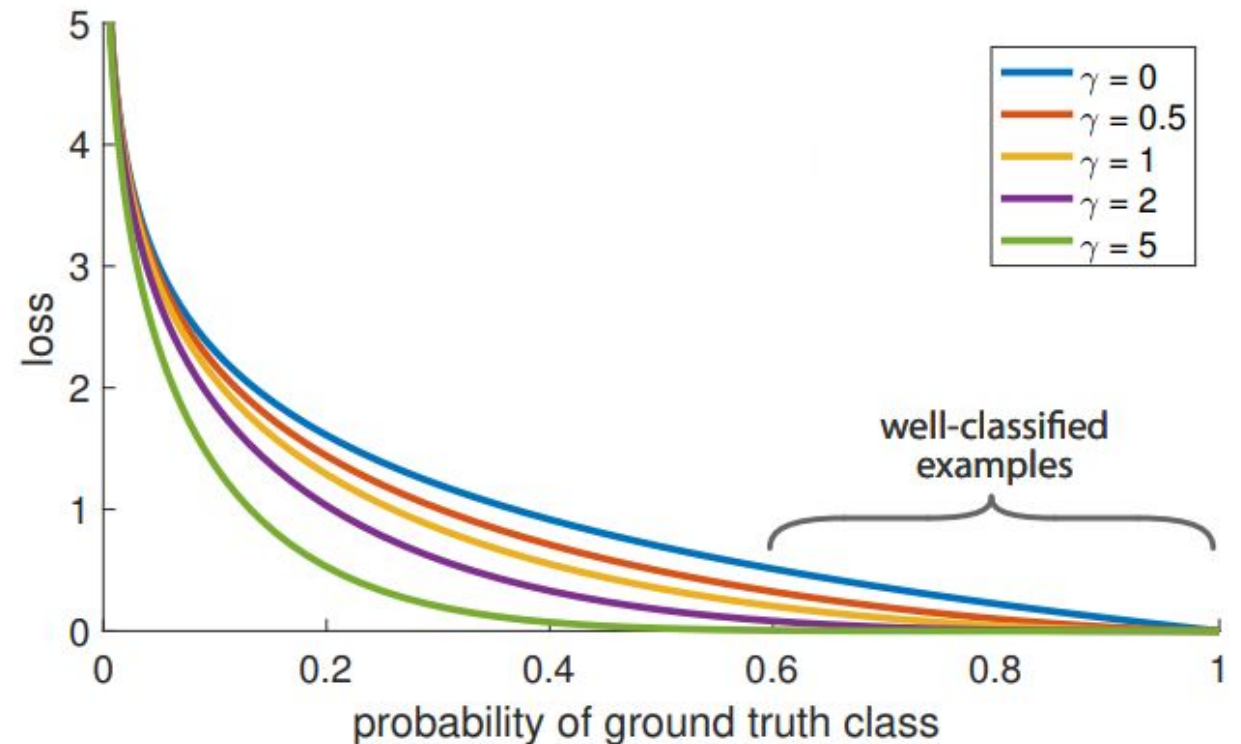
Focal Loss

Focal Loss: parameter gamma determines loss weight:

$$\text{CE}(p_t) = -\log(p_t)$$

$$\text{FL}(p_t) = -(1 - p_t)^\gamma \log(p_t)$$

logarithm in cross entropy CE is replaced
with weighted logarithm (FL)



Focal Loss

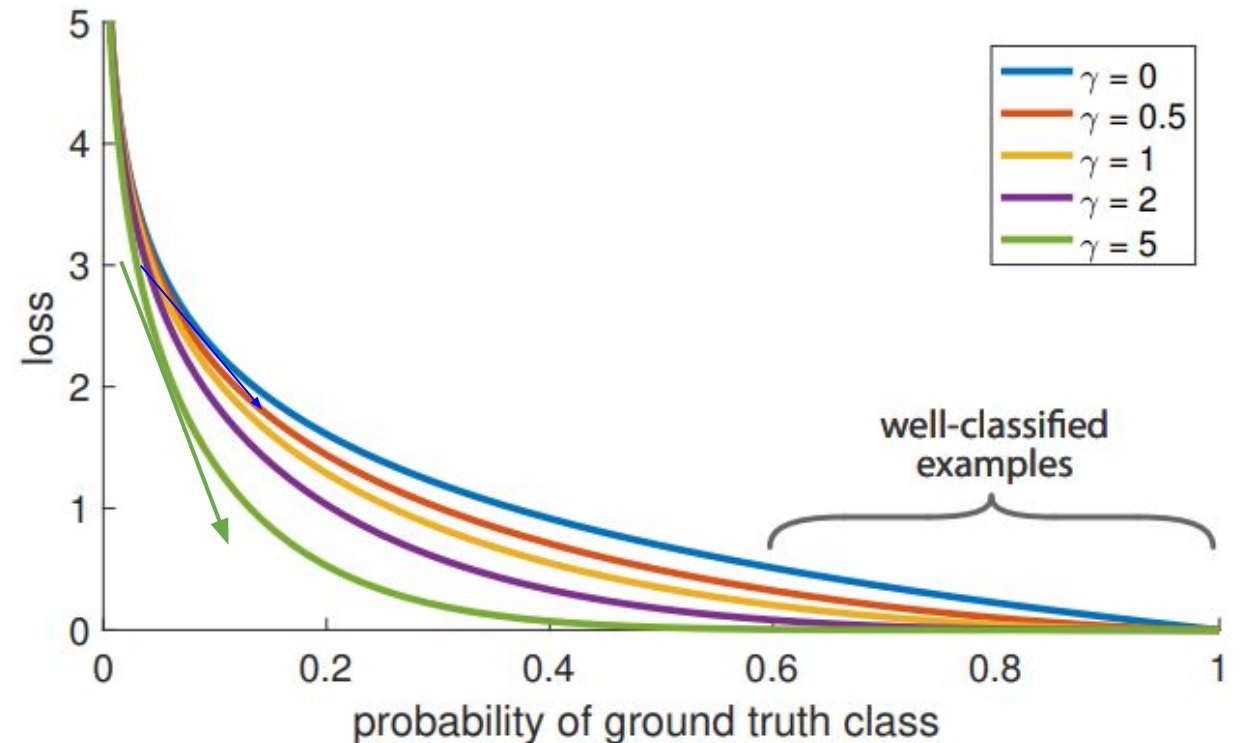
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$$\text{CE}(p_t) = -\log(p_t)$$

$$\text{FL}(p_t) = -(1 - p_t)^\gamma \log(p_t)$$

logarithm in cross entropy CE is replaced
with weighted logarithm (FL)

Stronger gradient for bad classification
Weaker gradient for good classification



Agenda for next time

Semantic segmentation and architectures:

Basis for many different computer vision tasks.

Agenda for next time

Segmentation / Scene parsing tasks continued:

- Object detection: locate individual objects in the image (with bounding box)
- Instance segmentation: predict masks for individual objects

Image-to-image tasks: transform image into another image

- Denoising
- Super-resolution

Questions!
